

Given the File Being Changed, What Else Should Be Read?

Import Structure, Issue Text and What the Issue Gives Away

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An agent editing one file of an unfamiliar repository must choose what else to read: the median pull request here is made against 286,385 tokens of code, of which the files it needs are 2.98%. Given the file being changed, we compare two signals for that choice — proximity in the import graph and the words of the issue report — and their fusion, on 2,412 change tasks from 1,162 pull requests in 112 Python repositories. Each ordering is scored by how much of an answer key it covers per line read, against two keys we never merge: the files co-edited in the pull request, and the files defining symbols the change uses.

Two findings survive our controls. First, what the issue is worth depends on whether it names a file the change touches. In 40.2% of tasks it names one as code; there the issue is worth 0.156 AUC to the best fusion, against 0.016 elsewhere. Deleting from the same issue only what points at the files — their names, then their directories, then the symbols they define — accounts for 0.061 to 0.078 of that 0.140 difference, and the rest survives every redaction; on seven repositories outside the sample the names alone mattered less, and not significantly on the co-edited key. A benchmark built from issue reports therefore averages an easy regime with a hard one, and a gain from issue-aware retrieval that is not reported per regime cannot be interpreted. Second, fusing a length-aware personalised walk with issue-to-path matching is the robust choice: on the pull requests that carry both keys it is never significantly worse than the best single signal under any stratification, and it held on seven repositories outside the sample, scored against predictions committed in advance. The exception is single-file changes, where the walk alone is better. The two-hop import neighbourhood, the usual argument for graph retrieval, holds 87.5% of the needed files and also half the repository.

Additional Key Words and Phrases: code search, context retrieval, dependency graphs, software maintenance, empirical software engineering

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1 INTRODUCTION

An agent is handed a bug report and a file to change. Before it can write anything it must decide what else to read, and it cannot simply read everything. In the corpus we study the median pull request is made against a repository of 286,385 tokens — counting the Python files in the import graph, excluding tests and non-Python files — and 74.8% of pull requests are made against a repository larger than a 128,000-token window, 60.5% against one larger than 200,000. Meanwhile the files that carry the answer are a median 2.98% of the repository. The reduction available is roughly 34×, and every line spent on an irrelevant file is a line not spent on a relevant one. Something has to choose, and the choice is the subject of this paper.

The question is narrower than an agent’s whole problem. Every task here hands the reader the file being changed; the question is what *else* to read. We measure retrieval, not repair — no agent is run — and the corpus comes from benchmarks that select for reproducible, test-verified fixes. Each of these is stated again where it bears on a result, but together they mean this paper answers “given the right file, what else belongs in the context”, not how well an agent that must also find that file will do.

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A human faces a softer version of the same problem — attention rather than tokens — and we return to that reading in Section 5. The budget is what makes the question sharp, so we pose it in those terms throughout.

The dominant answer in tool building is the dependency graph. Aider builds a repository map from the call and definition graph; RepoGraph, LocAgent and CoSIL construct code graphs and walk them outward from a starting point. The premise is intuitive and rarely stated: what you need to read is near, in the graph, to what you are changing.

There is a second signal, and it is the one a human reaches for first. Asked to fix the login page, a developer thinks of the authentication files — not because `login.py` imports them, which it may well not, but because the words match. That signal sits in the issue report.

We are not the first to notice it. SWE-bench’s own sparse-retrieval baseline is BM25 with the issue as the query [10], Agentless [22] leans on file and directory names, and the agentic systems all read the issue text as part of their loop. What is missing is not the idea that words help. It is a measurement of how far they carry against structure, on a stated answer key, under a stated budget, and with the issue corpus’s habit of naming the answer controlled for. Every system in Table 1 reports end-to-end accuracy against a single key, and none reports that comparison. That issues often name the file to change is known: Agentless’s manual pass over SWE-bench Lite [22] found that around half of its issues give the location of the file to edit, and models can name the buggy file from the issue alone often enough to raise questions of memorisation [13]. What has not been measured is how much of a retrieval method’s score that accounts for, which is what a benchmark average hides.

This paper compares the two, and the comparison needs a definition of which files a change needed. We use two that can be recovered without human labelling: the files *co-edited* in the same pull request, and the files that *define symbols* the changed code references; and we check both against a third, annotated by people, on a separate set of tasks. Graded as they come, the two keys rank the methods differently. Most of that difference is not about the signals. A symbol key exists for a single-file change and a co-edited key does not, so the two are scored on different pull requests; and in two tasks in five the issue names a changed file outright, which a method matching issue words against file paths can simply read. Every comparison that follows is therefore also reported on the pull requests that carry both keys, and within strata of how explicitly the issue names its files, with the repository as the statistical unit.

With those controls in place two findings remain, and the paper is organised around them. The first concerns the benchmark more than the methods: issue-aware retrieval is worth far more on issues that name their files, and a within-task redaction experiment measures how much of that is the names themselves. We call the names *leakage* only in this benchmark sense. In deployment an issue that names `core.py` is a legitimate input; the problem is that an average over issues that do and issues that do not describes neither, and the share of each varies by corpus. The second finding is a recommendation: fusing a walk from the file with issue-to-path matching is robust across both keys, every stratum, an external set of repositories and a human-annotated key, and it survives the loss of the free seed — except for single-file changes, where the walk alone is better. Whether the walk should favour short files depends on how a context is paid for, and we show both cases.

Research questions

RQ1 How much of a repository is the seed’s two-hop neighbourhood, and how much recall does it buy over a same-sized random sample?

RQ2 Do structural and lexical reading orders rank differently under the two evidence sources, and by how much?

RQ3 How much does conditioning on the seed file add over a query-independent repository prior, and does the answer depend on the evidence source?

RQ4 Do the two signals retrieve the same files, or different ones?

RQ5 Does fusing them produce an ordering that is good under both evidence sources?

RQ6 What fraction of tasks could be solved at all under a given budget — that is, how far is the best ordering we test from one that cannot be improved on?

RQ7 How much of what issue text contributes is the issue naming the answer? When an issue names a key file as code, and when it does not, what does an ordering gain from reading the issue at all?

RQ8 Do the answers survive a different cost rule, a human-annotated key, a dense retriever, and the loss of the free seed?

RQ7 and RQ8 were not in the design freeze; RQ7 grew out of the leakage control, which was, and RQ8 out of review. Section 3 lists every analysis added after the freeze, and which of them were tested against predictions committed in advance.

Contributions

- (1) **A decomposition of what issue text is worth.** An issue-free twin of each fusion gives the issue's total value, and three nested within-task redactions give the part carried by what points at the key files. On issues that name a key file as code the issue is worth +0.156 AUC to the best fusion, of which 0.061 to 0.078 is attributable to the names, depending on how much beyond the filename is deleted (Section 4.14).
- (2) **A robust recommendation, tested three ways.** Reciprocal-rank fusion of a personalised walk with issue-to-path matching is never significantly worse than the best pure ordering on the matched population. It held under predictions committed before scoring on seven unseen repositories (where one of four predictions, about redaction, failed) and against ContextBench's human-annotated key (all four held), and it remains the best ordering when the file to change is not given but guessed from the issue (Section 4.15). Whether the walk should be length-aware depends on how reading is charged: by lines it should, and two fusions differing only in that are 0.042 and 0.050 AUC apart; charged per file, it should not (Section 4.8).
- (3) **Three controls for this kind of comparison**, each of which changed a conclusion here: stratifying by whether the issue names a key file as code (40.2% of tasks do); matching the populations two keys are scored on (a co-edited key needs a multi-file change); and taking the repository as the unit, which leaves the headline comparisons significant but changes the significance verdict on 104 of 1,300: 102 of the 1,076 comparisons significant with the pull request as the unit lose significance, 2 gain it, and none changes sign.
- (4) **Three corrections to common arguments for graph retrieval.** The two-hop neighbourhood holds 87.5% of the needed files and 56% of the repository, at a median precision of 3.4%. Direction matters in one way only: walking against import edges is costly, while discarding direction costs nothing measurable. And the advantage structure shows on a key derived from imports is not corroborated by a human-annotated one: on the context a fix reads but does not edit, which is what a walk is claimed to find, walks and issue text are indistinguishable (Section 4.11).
- (5) **A cost accounting and a replication package.** Python source runs to 8.50 tokens per line; an oracle fits the whole key within 8,000 lines for 97.5% of pull requests. Given its

answer keys, the package rebuilds the study from a clean checkout and reproduces every committed score exactly; rebuilding the symbol keys themselves on another machine changes 41 of 1,886 and no conclusion (Section 7).

2 BACKGROUND AND RELATED WORK

Dependency structure as a guide to change. Using program structure to decide what a change touches is old. Impact analysis [4] propagates from a changed element along dependencies; Robillard [20] ranks program elements by the topology of their dependencies; feature-location surveys [7] catalogue structural, textual and dynamic techniques and note that they are usually evaluated in isolation. Ko et al. [11] observed developers navigating largely by identifier names and search rather than by following dependencies.

Change history. Zimmermann et al. [28] and Ying et al. [25] mine co-change from version histories to suggest files that change together. Our co-edited evidence source is the same notion of relatedness, used as ground truth rather than as a method.

Graph-based context for agents. SWE-bench [10] and its successors [3, 26] made repository-scale change tasks a standard benchmark, and a family of systems now assembles context by walking a code graph: Aider’s repository map [1], RepoGraph [16], LocAgent [6] and CoSIL [9]. Agentless [22] is a notable counterexample that leans on file and directory names.

What these systems share is the ground truth they are judged against. Table 1 records it for every system whose evaluation section we read in the published PDF; each entry is quoted or paraphrased from that section, not from an abstract or a secondary description. Where localization accuracy is reported at all, the target is the set of files or functions touched by the gold patch — the SWE-bench convention, and exactly our `co_edited` source. That is the evidence source on which, in our results, structural proximity is the weaker of the two signals on the uncontrolled comparison — an ordering that the leakage control of Section 4.13 then removes.

Leakage from the issue text. That a benchmark’s issue reports can contain their own answers is known. SWE-Bench+ [2] finds that 32.67% of the patches SWE-Agent with GPT-4 gets *passed* had the solution already present in the issue report or its comments, and Agentless [22] reports a manual pass over SWE-bench Lite that turns up problems whose issue description contains the exact ground-truth patch. Both concern the *solution*: the code, or a direct statement of the fix.

A weaker form concerns only *where*. A retrieval component does not need the fix; it needs only to know which files to open, and an issue that says `core.py` has supplied that without supplying anything else. This too is known. The same Agentless study classifies the location information in SWE-bench Lite’s issues and finds that around half already provide the location of the file to be edited; and the SWE-Bench Illusion study [13] shows that models can name the buggy file from the issue alone, without seeing the repository, in up to 76% of SWE-bench Verified tasks against 53% on repositories outside SWE-bench. Our own rate, 40.2% of tasks naming a key file as code and 12.6% quoting a full path, is in line with the first. What we add is not the rate but its consequence for retrieval measurement: how much of an issue-aware method’s score it accounts for, separated within task from the rest of what the issue says (Section 4.14). None of the systems in Table 1 reports its results by this stratum, and the practice we argue for in Section 4.13 is to do so as routinely as reporting the corpus size.

Concurrent work is questioning the key. The patch convention is under active revision, by at least four efforts that appeared while this study was running. SWE-Explore [27] derives line-level ground truth “from independent agent trajectories that successfully solved the same issue, distilling the specific code regions their solution paths actually consulted”, and scores a ranked list

Table 1. What repository-context systems are graded against, read from each paper’s own evaluation section (2026-09-23). The upper block is the patch-based convention; the lower block is the concurrent work that departs from it.

System	Venue	Ground truth used for evaluation
<i>Files touched by the gold patch</i>		
LocAgent [6]	ACL 2025	“The affected files or functions ...as identified in the patches, are considered the target locations”
CoSIL [9]	ASE 2025	File- and function-level accuracy on SWE-bench Lite and Verified, i.e. the patched locations
Agentless [22]	FSE 2025	Correct if the edit covers a superset of the ground-truth patch locations, at line, function and file level
LingmaAgent [15]	FSE 2025	Fault locations extracted from the developer patch, at file and function level
SWE-Fixer [23]	ACL 2025	Precision and recall against the defective files of the resolving pull request; BM25 top-30 then a learned reranker
KGCompass [24]	2025	Files and code elements modified by the ground-truth patch, scored by Jaccard overlap
SWE-Adept [8]	2026	Correct only if <i>all</i> locations modified by the ground-truth patch appear in the top- <i>k</i>
SWERank [18]	2025	Every function modified in the pull request is a positive
CodeScout [21]	2026	Reuses LocAgent’s patch-processing scripts: modified files, modules and functions
RepoGraph [16]	ICLR 2025	“The percentage of problems where the edit locations match the ground truth patch”, at file, function and line level
<i>Something other than the gold patch alone</i>		
GraphLocator [14]	2025	Locations extracted from the fix patch, then dual-annotated and cross-reviewed because patch-derived ground truth “contains missing locations”
SWE-Explore [27]	2026	File and line intervals that successful agents actually <i>read</i> , intersected across several verified trajectories
ContextBench [12]	2026	Human-annotated context traced outward from the patch regions, pruned by a second annotator, validated by an LLM patching from it alone
AgentRetrievalBench [17]	2026	A file is relevant if reading it helps the agent’s next step; one of four task types is ripple files beyond an anchor
CodeR [5]	2024	End-to-end on SWE-bench, but its separate localization study uses golden locations marked by the authors

“under a fixed line budget” — both the key and the cost model are close relatives of ours. ContextBench [12] annotates gold context by hand, 522,115 lines over 1,136 tasks, and measures recall, precision and efficiency rather than end-to-end resolution. AgentRetrievalBench [17] defines a file as relevant “when reading it would materially help the agent take the next correct step”, evaluated under a top-*k* or token budget; one of its four task types, ripple files beyond an anchored edit, is our co-edited source by another name. And GraphLocator [14] keeps the patch key but dual-annotates it, on the stated grounds that patch-derived ground truth misses locations.

We take this as corroboration of the premise rather than as a threat to it: several groups independently concluded in the same year that the patch key is inadequate. What none of them reports is the consequence we measure. Each proposes a better key and evaluates methods against it; we hold two keys already recoverable from existing benchmark data, and show that the choice between them changes the ranking of the methods themselves. The contribution is not the observation that keys differ, and it is not the reversal either: Sections 4.3 and 4.13 show that both halves of that reversal are produced by confounds rather than by the signals. The contribution is the controls

that establish this, and the measurement in Section 4.14 of how little issue text adds once the tasks whose issues name the answer are set aside.

Retrieval. BM25 [19] remains the reference lexical ranking function; we use it unchanged, varying only the query.

3 STUDY DESIGN

The corpus, the graph builder, the task definition and the coverage–cost metric were fixed and tagged in version control before any method was run; the present study adds method families and analyses on top of that frozen base. Changes made after that tag, and claims we have retracted, are recorded in the replication package rather than silently absorbed.

The freeze covers the corpus, the graphs, the task definition and the metric. It does not cover the method families: the fusion orderings, the directed walks and the lexical methods queried by issue text were all added afterwards, in response to what the first results showed. Fusion is the paper’s main recommendation and it was an *exploratory* result when we first found it. We then tested it twice under predictions committed before any data were scored: on seven repositories outside the sample (Section 4.10) and against a human-annotated key (Section 4.11). Three later additions, made in response to review, were not fixed in advance and are reported as exploratory: the two alternative cost rules (Section 4.8), a dense retriever (Section ??) and a condition without a free seed (Section 4.15).

We also use “design freeze” rather than “pre-registration” throughout, and mean the weaker thing. The freeze is a signed tag in our own version control, not a record lodged with an external registry, so it establishes the order in which we did things to a reader who trusts the repository and to nobody else. The protocol in Section 6 has the same status. We report it because stating a design before running it is worth something even unregistered, not because it carries the guarantee that an OSF entry would.

3.1 Repositories and tasks

We searched SWE-bench, SWE-bench-Live, SWE-Gym, SWE-bench Pro, Loc-Bench and SWE-rebench for candidate repositories, and drew the sample’s pull requests from three of them — SWE-bench, SWE-bench-Live and SWE-rebench — keeping Python repositories that pass the sampling rules; the repositories whose tasks come only from SWE-Gym or SWE-bench Pro form the external set of Section 4.10. A pull request that edits k existing non-test source files that are nodes of the import graph yields k tasks: each edited file in turn becomes the *seed*, and the task is to order the remaining files of the repository. This gives 3,123 tasks from 1,873 pull requests in 113 repositories. Of these, 2,412 tasks — 1,162 pull requests in 112 repositories — have at least one key file inside the graph once the seed is removed, and they are the ones scored here; the remaining 711 tasks have an empty key under both sources and are reported but not scored. Those 711 tasks come from 711 distinct pull requests, and 706 of them change a single file: with no second edited file there is no co-edited key, and the symbol key is empty whenever the changed code resolves to nothing outside itself — standard library, third-party packages, or names `jedi` could not resolve. That is 61.2% of all single-file tasks, which is a real limitation of the symbol key and not a filtering choice of ours. Within the scored set, 1,958 tasks (708 pull requests) have a co-edited key and 2,341 (1,156 pull requests) a symbol key.

The median task’s repository has 131 graph nodes, and the median key is 2 files under either source — 3 files totalling 1,515 lines if the two are pooled, as they are only in Table 3 — so the task is to find a handful of files among roughly a hundred.

3.2 Import graphs

Graphs are built per base commit with a custom ast walker over the repository at that commit. Nodes are source files; an edge $a \rightarrow b$ means a imports a module resolving to b . Package `__init__` files are not inserted as intermediaries, from `X import Y` resolves to the module, edges are unweighted, imports inside `TYPE_CHECKING` blocks and inside function bodies are kept, and test files are excluded. Per-file line counts come from the same pass and are the cost unit throughout.

3.3 Two evidence sources

co_edited The other non-test source files changed by the same pull request. This is what a developer in fact touched, and it is independent of the import graph.

symbol Files that define names the changed code references, resolved with a static analyser. This is what the change depends on.

Symbol resolution uses Jedi's `goto` with `follow_imports=True`, which reaches a name's defining file by traversing import statements. This bounds what the symbol key can show: the key is constructed by walking the import graph, so a method that walks the import graph is being scored against a target built from the same substrate. We report it because it is the natural static notion of "what this code depends on", not because it is a neutral arbiter between the two method families. Names are resolved in the code with the pull request's patch applied, from the lines it adds, and a defining file must already exist at the base commit. A definition the fix itself adds can therefore put a file in the symbol key, but only a file the fix modifies, which is also in the co-edited key; 47.5% of symbol-key file references are, which bounds that effect from above.

By *issue text* we mean the problem statement only; the `hints_text` some benchmarks ship, written after the fact by maintainers, is not used.

The seed file is removed from both. We deliberately do not merge them: the whole question is whether the choice matters. It does, and the union is not a neutral compromise — the symbol key contributes 7,675 file references against 4,458 for co-editing, so a union largely inherits the structural source's verdict.

A third source — files actually executed by the tests that the fix repairs — is defined in our protocol and was collected for 113 pull requests before we stopped. We exclude it rather than report it: a source covering under a tenth of the corpus cannot be compared against two that cover all of it, and reporting it alongside them would invite exactly the conflation this paper argues against. The partial collection is in the replication package so that it is visible rather than quietly dropped, and completing it is the obvious extension.

3.4 Reading orders

Every method receives the graph at the base commit, the seed file, per-file identifier bags and the issue text, and returns an ordering of all other nodes. Table 2 lists them by family. Identifier bags are built by splitting every identifier in the file on underscores and camel-case boundaries and lower-casing; the issue text is tokenised identically, so text is matched against code over one vocabulary. BM25 is used with the standard $k_1 = 1.5$, $b = 0.75$ and differs between methods only in the query. Fusion is reciprocal-rank fusion with $k = 60$.

Every issue-aware ordering in Table 2 is lexical. To test whether a learned representation extracts more from an issue than BM25 does, we also score a dense retriever, added after the main results (Section ??). It uses the embedding model fixed in our design before any method ran, `jina-embeddings-v2-base-code`, on the first 512 tokens of each file; an issue is embedded whole, as the length-weighted mean of its 512-token chunks, so that nothing in it is truncated. Files are ranked by cosine similarity to the issue (`dense_issue`) or to the seed file (`dense_seed`), and the

Table 2. Reading orders, by family. Structural methods use the seed and the graph; global methods use neither the seed nor the issue; lexical methods use no graph.

Family	Method	Ordering
structure	bfs_und	breadth-first from the seed, undirected
	hops_lines	by (hop distance from seed, file length)
	ppr_und	personalised PageRank from the seed, undirected
	ppr_und_pl	the same, divided by file length
	ppr_out	the same walk along import edges: what the seed depends on
	ppr_out_pl	the same, divided by file length
	ppr_in	the same walk against import edges: what depends on the seed
global	ppr_in_pl	the same, divided by file length
	pagerank indegree	PageRank of the whole graph; ignores seed and issue number of files importing this one
lexical	bm25_seed	BM25, query = identifiers of the seed file
	path_seed	BM25 over paths, query = the seed's path tokens
	bm25_issue	BM25, query = identifiers of the issue text
	bm25_issue_pl	the same, divided by file length
	path_issue	BM25 over file paths, query = issue tokens
fusion	rrf_hops_path	RRF of hops_lines and path_issue
	rrf_hops_issue_path	RRF of hops_lines, bm25_issue, path_issue
	rrf_pprpl_issue_path	RRF of ppr_und_pl, bm25_issue, path_issue
	rrf_ppr_issue	RRF of ppr_und and bm25_issue
	rrf_ppr_issue_path	RRF of ppr_und, bm25_issue, path_issue
reference	same_dir	files in the seed's directory first
	random	uniformly shuffled: the floor
	oracle	key files only, cheapest first: the ceiling

dense ranking is fused with the length-aware walk in place of BM25 over contents or beside it. These orderings are scored but kept out of the ranks and correction families of the other tables, so that adding them moves nothing reported elsewhere.

Two asymmetries matter. The structural methods are *given* a correct answer for free — the seed is one of the files the pull request changed. The issue-text methods are given no seed at all. And the metric charges by lines read, so we include length-aware variants (suffix _pl, score divided by file length) to separate “ranks the right files highly” from “happens to prefer short files”.

3.5 Metric

An ordering is read from the top, whole file by whole file, until the next file would exceed the line budget; reading stops there, so no file is skipped to fit a smaller one further down and none is read in part. Coverage is the fraction of the key that has been read. We evaluate at budgets {250, 500, 1000, 2000, 4000, 8000, 16000, 32000} and report the AUC, the mean coverage over those eight doublings, so that a method must be good at both tight and generous budgets. We also report the more directly interpretable quantity: the number of lines that must be read before half the key has been covered.

Orderings with ties are re-drawn three times per task with distinct seeds and the coverage curves averaged.

The upper budgets approach the whole repository for the smaller repositories, which could compress differences between methods. Restricting the AUC to the six budgets up to 8,000 lines leaves the leading orderings unchanged under both keys and on both populations.

The budget is denominated in lines because the pre-registered design fixed it that way at freeze-1, and we do not change a frozen metric after seeing results. Lines are nevertheless the wrong unit for a context window, so we measure the conversion rather than assume it: across the 121,075 distinct file versions in the corpus, tokenised with the `cl100k_base` byte-pair vocabulary, Python source runs to 8.50 tokens per line in aggregate and a median of 7.88 per file, with a 5th–95th percentile range of 5.70–10.90. Absolute counts shift by a small constant factor between tokenizer families; the ratios we report do not depend on the choice. A budget of 8,000 lines is therefore roughly 68,000 tokens, and the eight budgets span roughly 2,100 to 272,000 tokens — from a fraction of a small context window to more than a large one.

3.6 Statistics

Task AUCs are averaged within a pull request first, so a change touching many files does not count many times. Paired comparisons use the Wilcoxon signed-rank test with the Vargha–Delaney A_{12} effect size, Holm-corrected within each family of comparisons. Confidence intervals are 2,000-resample bootstraps resampling *repositories*, not tasks, so that a few large repositories cannot manufacture significance.

Every rank in this paper is on one scale: position among the twenty-three orderings we score, the oracle included, so the best attainable non-oracle rank is 2. Where we say “the twenty contenders” we mean the subset entered into the paired tests — the twenty-three less the oracle, indegree and `bfs_und`, which are reported for reference but not tested. Two further orderings, the issue-free fusion twins of Section 4.14, are scored but never ranked: they are controls rather than candidate reading orders, and ranking them would shift every other rank in every table.

The paired tests are reported with the **repository** as the unit: the pull-request scores are averaged within a repository and the test runs over the 112 repositories. Pull requests are not independent draws — they cluster in repositories that differ in size, layout convention and issue-writing culture — and a test that ignores the clustering counts the same evidence many times. The difference is not cosmetic, though it does not touch the headline comparisons, which are significant either way. It touches the borderline ones: of the 1,076 paired comparisons on the two keys that are significant after Holm correction with the pull request as the unit, 102 are not with the repository as the unit, and 2 move the other way; no estimate changes sign. One of the 102 is a comparison this paper would otherwise have leaned on — `hops_lines` against `path_issue` on co-edited files, $p_{\text{Holm}} < 10^{-4}$ over pull requests and 0.07 over repositories. We report the pull-request-level tests alongside, in the replication package, so the two can be compared.

Averaging within a repository has a cost of its own: a repository that contributes one pull request weighs as much as one that contributes a hundred. So every comparison a claim rests on is also re-estimated as a linear mixed model of the per-pull-request paired difference, with a random intercept per repository (Table 22). It keeps every pull request as an observation and models the clustering instead of averaging it away. On 40 of those 41 comparisons it reaches the same verdict, significant or not, as the repository-level Wilcoxon test. The exception is the borderline comparison above: weighting repositories by their pull requests, the mixed model finds `path` matching better than `hops_lines` on co-edited files by 0.044 ($p_{\text{Holm}} < 10^{-3}$), where the repository-level test does not. We report it, and no claim in this paper rests on that comparison being null.

Table 3. Where the key files sit relative to the seed, and how big the two-hop ball is. Hop shares are pooled proportions over all key-file references. The ball rows are computed against the union of the two keys over 2,412 tasks: files and lines are medians, the share of the repository and the recall are means, and precision is the median per task.

Key files at hop distance	1	2	3+	unreachable
co_edited	39.0%	49.1%	9.1%	2.8%
symbol	45.3%	41.4%	9.1%	4.2%
<i>The two-hop ball: 68 files, 21,646 lines, 56% of the repository</i>				
recall 87.5%, precision 3.4%, recall lift over a same-sized random sample 1.57×				
<i>The union key: 3 files, 1,515 lines</i>				

Holm correction is applied within families that are fixed by the design rather than chosen per claim. Table 21 lists them: each set of paired tests for one key, one unit, one population and one leakage stratum is one family of 325 tests, among twenty-six orderings — the twenty contenders, the two issue-free twins and the four names-redacted variants of Section 4.14, so that every twin and redaction comparison has its test; each redaction arm is one family; and the mixed-model re-estimates are one family.

Because a co-edited key requires a pull request that touches more than one graph file while a symbol key does not, the two keys do not cover the same pull requests. Wherever we compare across keys we therefore report the comparison twice: over each key’s full population, and over the 702 pull requests that carry both (Section 4.3).

4 RESULTS

4.1 RQ1: the two-hop neighbourhood is half the repository

“The files you need are within two hops” is true here: 86.7% of symbol-key files and 88.1% of co-edited files lie within two undirected hops of the seed (Table 3). Taken alone this reads as strong evidence for graph retrieval.

The other half of the statistic undoes it. The median two-hop ball holds 68 files and 21,646 lines, more than five times a 4,000-line budget. Measured task by task against its own repository, the ball is a median 58.5% of it and a mean 55.6%. Its median precision is 3.4%. Against a random sample of the same size it lifts recall by a factor of 1.57.

The two-hop neighbourhood is therefore not a filter. It is a statement that the files you need are somewhere in about half the repository — which is exactly the regime in which the *ordering* decides everything and the graph has already said all it has to say.

4.2 RQ2: the two keys rank the methods differently

Table 5 is the uncontrolled comparison, and the one whose reading the rest of this section revises. Throughout the paper, the AUC in a table is a mean over pull requests, while a difference quoted in the text is the mean of the paired per-repository differences, because the repository is the unit every test uses (Section 3.6). The two agree to within a few thousandths and can differ in the third decimal. Under the co-edited key the five best non-oracle orderings are four fusions and one lexical method; the best purely structural ordering is tenth, and the query-independent priors are twenty-first and twenty-second of twenty-three, below `same_dir` and barely above random. Under the symbol key the table inverts: the four best non-oracle orderings are three structural orderings and one fusion, and the best lexical ordering is fifteenth.

The individual reversals are large and significant after Holm correction with the repository as the unit. Issue-text path matching beats length-aware personalised PageRank by 0.059 AUC on co-edited files ($A_{12} = 0.403$ for the walk, $p_{\text{Holm}} < 10^{-4}$) and loses to it by 0.100 on symbol files ($A_{12} = 0.671$, $p_{\text{Holm}} < 10^{-7}$).

We rest the reversal on those paired tests rather than on a rank correlation. The correlation is low – Spearman $\rho = 0.24$ over the twenty contenders ($p = 0.31$), and $\rho = 0.37$ over all twenty-two non-oracle orderings ($p = 0.09$) – but with twenty points a non-significant ρ is weak evidence of anything, and its value moves with which methods are included. The Holm-corrected pairwise reversals do not have that weakness.

Two things about Table 5 are *not* safe to read off it, and the next two subsections are about them. The two columns are computed over different pull requests, and the left column is computed over a corpus in which many issues name the answer outright. Each of those turns out to account for one of the two halves of the reversal.

Figure 1 shows the same reversal as coverage curves. Table 6 states it in the unit a context budget is actually spent in: to reach half the co-edited files a reader following length-aware personalised PageRank consumes a median of 6,155 lines, against 4,838 following issue-text path matching and 3,342 following the best fusion. On symbol files the same three numbers are 3,483, 6,638 and 3,957 – the walk and the words trade places, and the fusion is close to the better of the two in both.

4.3 The reversal is smaller on a matched population

A co-edited key exists only for a pull request that changes more than one file in the graph. A symbol key does not need that, and 454 of the 1,156 pull requests with a symbol key change a single file. The two columns of Table 5 are therefore scored on different samples, and single-file changes are exactly the case where a seed’s import neighbourhood is most of what there is to find. The comparison has to be repeated on a population that carries both keys.

Table 4 does that, on the 702 pull requests that have both. On the co-edited key nothing moves: no method changes rank by more than one place. On the symbol key the structural family loses most of its advantage. Cost-scaled personalised PageRank along import edges falls from second to eighth (0.491 to 0.417), the undirected form from third to fifth, and hops_lines from fourth to sixth, while issue-to-path matching rises from fifteenth to ninth. The three top places under both keys are then held by the same three fusions.

The statistical claim moves with the ranks. On the full corpus a walk along import edges beats path matching on the symbol key by 0.101 ($p_{\text{Holm}} < 10^{-7}$, repository as the unit). On the matched population the same comparison gives +0.008 and $p_{\text{Holm}} = 1$; for hops_lines against path matching it gives +0.019 and $p_{\text{Holm}} = 1$. We therefore do not claim that structural proximity beats issue-text retrieval on the symbol key. On a like-for-like sample the two are indistinguishable, and what remains of the reversal is the co-edited half – which the next control also qualifies.

4.4 Direction: reversing a walk is costly, discarding it is free

The walks above are undirected, as they are in most of this literature. An import edge has a direction, so we add both orientations of the personalised walk, length-aware and not.

The result is asymmetric, and the asymmetry is the finding. Running the walk *along* import edges – towards what the seed depends on – places ppr_out_pl second of twenty-three on the symbol key at 0.491. Running it *against* them collapses to ppr_in_pl at 0.277, twenty-first. The like-for-like gap is 0.123 on the matched population and 0.215 on the full one ($p_{\text{Holm}} < 10^{-14}$, repository as the unit), and on the symbol key the ten largest gaps between any two structural orderings, on either population, all have an inward walk on the losing side.

Table 4. The two keys re-scored on the 702 pull requests that carry both, top ten under each. Δ is the change in rank against the full population of Table 5; a positive number means the method does worse once the samples are matched.

Method	Fam.	AUC	#	Δ
oracle	ref	.774	1	–
rrf_pprpl_issue_path	fus	.487	2	–
rrf_hops_issue_path	fus	.480	3	–
rrf_hops_path	fus	.457	4	–
rrf_ppr_issue_path	fus	.445	5	–
path_issue	lex	.435	6	–
bm25_issue	lex	.409	7	–
rrf_ppr_issue	fus	.394	8	–1
bm25_seed	lex	.393	9	+1
hops_lines	str	.389	10	–
oracle	ref	.779	1	–
rrf_pprpl_issue_path	fus	.470	2	–4
rrf_hops_path	fus	.456	3	–2
rrf_hops_issue_path	fus	.454	4	–4
ppr_und_pl	str	.436	5	+2
hops_lines	str	.425	6	+2
rrf_ppr_issue_path	fus	.421	7	–4
ppr_out_pl	str	.417	8	+6
path_issue	lex	.404	9	–6
ppr_und	str	.395	10	+1

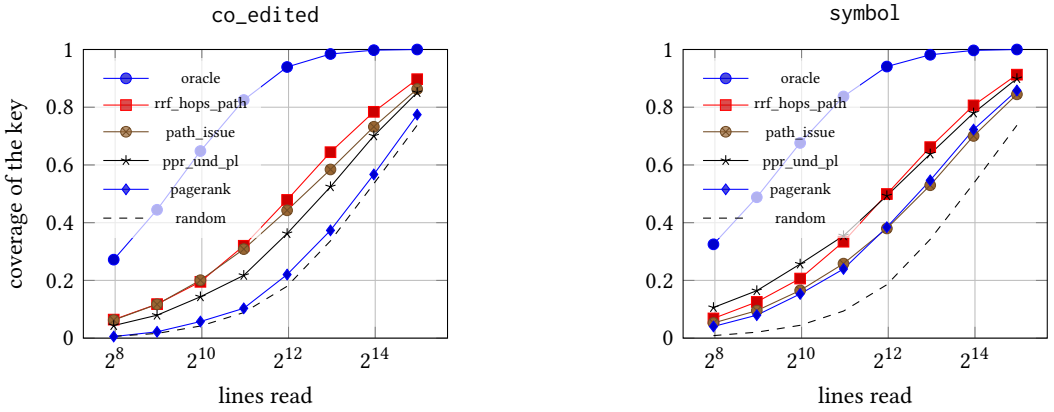


Fig. 1. Mean coverage against lines read, on the full corpus and without either control. The same five orderings change places between the two evidence sources: issue-text path matching tracks the best fusion on co-edited files and falls below the query-independent prior on symbol files, while length-aware personalised PageRank does the reverse. Sections 4.3 and 4.13 show that both halves of this crossing have an explanation that is not about the signals.

Table 5. Mean AUC by evidence source, pull request as the unit, with bootstrap 95% confidence intervals over repositories. The same twenty-three orderings, ranked twice.

co_edited (708 PRs)				symbol (1,156 PRs)			
#	Method	Fam.	AUC [CI]	#	Method	Fam.	AUC [CI]
1	oracle	ref	0.773 [.750,.796]	1	oracle	ref	0.793 [.772,.813]
2	rrf_pprpl_issue_path	fus	0.486 [.455,.517]	2	ppr_out_pl	str	0.491 [.463,.517]
3	rrf_hops_issue_path	fus	0.480 [.448,.510]	3	ppr_und_pl	str	0.489 [.460,.517]
4	rrf_hops_path	fus	0.457 [.425,.489]	4	hops_lines	str	0.477 [.447,.505]
5	rrf_ppr_issue_path	fus	0.445 [.414,.475]	5	rrf_hops_path	fus	0.476 [.449,.502]
6	path_issue	lex	0.434 [.405,.465]	6	rrf_pprpl_issue_path	fus	0.474 [.445,.501]
7	bm25_issue	lex	0.407 [.379,.435]	7	ppr_out	str	0.458 [.430,.487]
8	bm25_seed	lex	0.393 [.362,.425]	8	rrf_hops_issue_path	fus	0.457 [.429,.483]
9	rrf_ppr_issue	fus	0.393 [.363,.425]	9	ppr_und	str	0.446 [.417,.475]
10	hops_lines	str	0.389 [.354,.423]	10	bfs_und	str	0.420 [.391,.447]
11	ppr_und_pl	str	0.371 [.338,.403]	11	rrf_ppr_issue_path	fus	0.420 [.392,.448]
12	path_seed	lex	0.371 [.338,.404]	12	pagerank	glo	0.410 [.381,.439]
13	bfs_und	str	0.362 [.330,.395]	13	indegree	glo	0.398 [.371,.425]
14	ppr_und	str	0.357 [.325,.391]	14	rrf_ppr_issue	fus	0.390 [.361,.421]
15	bm25_issue_pl	lex	0.354 [.318,.389]	15	path_issue	lex	0.382 [.352,.412]
16	same_dir	ref	0.329 [.298,.360]	16	path_seed	lex	0.365 [.336,.392]
17	ppr_out_pl	str	0.327 [.296,.358]	17	bm25_issue_pl	lex	0.355 [.324,.386]
18	ppr_out	str	0.312 [.282,.344]	18	bm25_issue	lex	0.340 [.311,.368]
19	ppr_in_pl	str	0.310 [.279,.341]	19	bm25_seed	lex	0.336 [.305,.365]
20	ppr_in	str	0.308 [.278,.339]	20	same_dir	ref	0.320 [.290,.349]
21	pagerank	glo	0.271 [.240,.301]	21	ppr_in_pl	str	0.277 [.248,.303]
22	indegree	glo	0.271 [.240,.301]	22	ppr_in	str	0.269 [.242,.295]
23	random	ref	0.248 [.220,.277]	23	random	ref	0.248 [.220,.276]

Table 6. Median lines that must be read before half, then all, of the key is covered. Lower is better.

Method	to half the key		to all of it	
	co_edited	symbol	co_edited	symbol
oracle	524	440	1,150	1,349
rrf_pprpl_issue_path	3,342	3,957	7,213	9,563
rrf_hops_path	3,962	3,715	8,073	9,636
ppr_out_pl	8,655	3,924	15,396	12,386
ppr_und_pl	6,155	3,483	11,644	11,303
hops_lines	5,620	4,080	11,454	11,582
path_issue	4,838	6,638	11,072	14,504
bm25_issue	5,012	7,451	10,926	16,699
random	14,557	14,713	21,026	22,643

But the undirected walk loses almost nothing. On the symbol key `ppr_und_pl` scores 0.489 against `ppr_out_pl`'s 0.491, a paired difference of 0.001 that is not significant on the full corpus ($p_{\text{Holm}} = 1$) and reverses in sign on the matched one, where the undirected walk leads by 0.020 ($p_{\text{Holm}} = 0.85$). So the practice this literature has settled on — throw the direction away — is not the mistake it might appear to be from the spread of the table. Walking backwards is the mistake.

On the co-edited key direction is worse than useless. All four directed variants rank seventeenth to twentieth, below `same_dir`, a baseline that orders files by directory and consults no graph at all;

Table 7. How each key file is reached within the top k of the structural family (ppr_und, hops_lines) and the lexical family (bm25_issue, path_issue).

Source	k	both	structure only	words only	neither
co_edited	10	32.2%	11.4%	30.1%	26.4%
co_edited	20	50.9%	9.5%	24.4%	15.3%
co_edited	50	76.5%	6.1%	13.3%	4.1%
symbol	10	28.4%	23.4%	20.5%	27.7%
symbol	20	47.2%	20.0%	16.5%	16.2%
symbol	50	72.9%	11.5%	9.8%	5.9%

all four undirected walks rank tenth to fourteenth, above it. Committing to an orientation costs 0.040 to 0.064 AUC there ($p_{\text{Holm}} < 10^{-3}$).

The explanation is the same one that qualifies the symbol key throughout. That key is built by resolving names with jedi, which follows imports in exactly one direction, so a walk in that direction is measuring something close to the key's own construction. The co-edited key records a human decision about what belonged in one change, and no orientation of the import graph recovers it well.

4.5 RQ3: what the seed is worth depends on the source too

A query-independent PageRank prior — a ranking that never looks at the seed or the issue, and simply says which files matter in this repository — reaches 0.410 on symbol files against 0.491 for the best length-aware personalised walk, which does look at the seed. That is 83% of the personalised score in absolute terms, or 67% of its margin over a random ordering, which is the fairer of the two readings and the one the fourth recommendation in Section 5 quotes. It is far above random (0.248; $A_{12} = 0.758$, $p_{\text{Holm}} < 10^{-16}$). On that evidence source, much of what the personalised walk delivers is a repository prior.

Under the co-edited key the same prior scores 0.271 against a random floor of 0.248. The difference is significant but negligible ($A_{12} = 0.552$, $p_{\text{Holm}} = 0.002$), and the personalised walk beats it by 0.101 ($A_{12} = 0.666$). Knowing which file is being changed therefore matters a great deal for predicting what else was changed, and rather little for predicting what the code calls, where knowing the repository's hubs is nearly as good.

The opposite claim is easy to make from a single source: whether “the graph knows what you are working on” appears true or false is itself a function of the answer key.

4.6 RQ4: the two signals name different files

The reversal is not two methods disagreeing about the order of one list. At $k = 10$ the top-10 sets of personalised PageRank and issue-text BM25 have a mean Jaccard overlap of 0.187, and of personalised PageRank and issue-text path matching only 0.102; at $k = 50$ these rise to 0.508 and 0.412. For comparison, personalised PageRank and the query-independent prior overlap at 0.364 at $k = 10$: a seed-conditioned walk resembles the repository prior more than it resembles a lexical ranking.

Table 7 follows individual key files. At $k = 20$, 24.4% of co-edited files are found only by the lexical family and 9.5% only by the structural family; on symbol files the asymmetry flips (20.0% structure-only against 16.5% words-only). Under either source about a sixth of key files are found by neither family at $k = 20$, which is where the remaining headroom to the oracle lives.

4.7 RQ5: fusion is the only ordering that survives both sources

If the evidence source is not known in advance — and for a tool shipped to users it is not — the quantity to optimise is the worst a method does across sources. We report that two ways. Worst-case *rank* is easy to read but depends on the competitor set: adding our four directed walks reshuffled everything below them, and a criterion that moves when other people's methods are added is a poor one to recommend. Worst-case *shortfall* — the largest AUC gap to the best non-oracle ordering under either key — says the same thing in units nobody else's entries can change. Table 10 gives both and sorts by the second. The two agree on the family and disagree on its best member: `rrf_pprpl_issue_path` has the smaller shortfall (0.017 against 0.029) while `rrf_hops_path` has the better worst rank (5 against 6). We recommend the former, because the shortfall criterion, the matched population (Section 4.3) and the held-out splits (Section 4.9) all select it and only the rank criterion does not. Three of the five fusions beat every pure signal. The two that do not — `rrf_ppr_issue` and `rrf_ppr_issue_path` — are not distinguished by whether they include path matching, since four of the five do and one of those still fails. They are the two built on the unscaled `ppr_und`. The three that survive all fuse from a *length-aware* structural input: either `ppr_und_pl`, whose scores are divided by file length, or `hops_lines`, which breaks ties between equidistant files by preferring the shorter one. The two differ in how they use length and agree in that they use it at all.

One pair isolates that property exactly. `rrf_pprpl_issue_path` and `rrf_ppr_issue_path` fuse the same three lists and differ only in whether the walk's scores are divided by file length first; their worst-case ranks are 6 and 11, and the difference is +0.042 AUC on co-edited files and +0.050 on symbol files on the matched population ($p_{\text{Holm}} < 10^{-11}$, repository as the unit). `rrf_ppr_issue` differs from its length-aware counterpart in two ways at once, so we rest the claim on the clean pair rather than on the three-of-five tally. How much of this margin the cost rule itself creates is the subject of Section 4.8.

The insurance is close to free. Over the full corpus `rrf_hops_path` gains 0.067 AUC on `hops_lines` on co-edited files ($A_{12} = 0.602$, $p_{\text{Holm}} < 10^{-11}$) and is indistinguishable from it on symbol files (+0.003 as a mean of repository differences, −0.001 in the pull-request means of Table 5; $p_{\text{Holm}} = 1$); against `ppr_und_pl` it gains 0.087 ($p_{\text{Holm}} < 10^{-13}$) and loses 0.008 ($p_{\text{Holm}} = 1$). The lexical signal adds a large amount where structure is weak and costs nothing where structure is strong. Section 4.14 asks what, in that lexical signal, is doing the work.

4.8 How much of the ranking is the cost rule?

The frozen metric charges whole files by their lines and stops at the first file that would overflow the budget. Under that rule, dividing a score by file length is close to a knapsack heuristic for the metric itself, and a long file placed early costs twice: its own lines, and every later file that no longer fits. The length-awareness result could therefore belong to the rule rather than to the signals. We rescored every ordering under two other rules, neither fixed in advance. Under *skip*, a file that does not fit is passed over and reading continues, so an early long file costs only its own lines. Under *files*, every file costs one unit whatever its length, at budgets of 1 to 128 files — the accounting of an agent whose cost is the number of files it opens rather than their size. Table 8 gives the leading orderings under all three rules.

Under the skip rule the recommendation stands. The fusion is the best non-oracle ordering on the symbol key and 0.002 behind `rrf_hops_issue_path` on the co-edited key; over the full corpus it is 0.003 behind on the co-edited key and leads on the symbol key, where under the frozen rule the walks lead. But the margin that length-awareness buys shrinks by about sixty per cent: the clean pair of Section 4.7 separates by +0.015 and +0.021 instead of +0.042 and +0.050, still significant

Table 8. AUC (and oracle-inclusive rank) of the leading orderings under three cost rules, on the matched population. *Stop* is the frozen rule: whole files by lines, reading stops at the first file that would overflow. *Skip* passes over such a file and continues. *Files* charges one unit per file, at budgets of 1 to 128 files.

Ordering	co_edited			symbol		
	stop	skip	files	stop	skip	files
rrf_pprpl_issue_path	.487 (2)	.519 (3)	.595 (7)	.470 (2)	.517 (2)	.571 (5)
rrf_ppr_issue_path	.445 (5)	.505 (4)	.649 (2)	.421 (7)	.496 (4)	.616 (2)
rrf_hops_issue_path	.480 (3)	.522 (2)	.631 (3)	.454 (4)	.509 (3)	.595 (3)
rrf_hops_path	.457 (4)	.479 (5)	.488 (10)	.456 (3)	.492 (5)	.483 (9)
hops_lines	.389 (10)	.399 (10)	.381 (15)	.425 (6)	.447 (9)	.410 (16)
ppr_und_pl	.372 (11)	.379 (14)	.268 (21)	.436 (5)	.448 (8)	.319 (20)
ppr_und	.358 (14)	.396 (11)	.497 (8)	.395 (10)	.456 (7)	.531 (8)
path_issue	.435 (6)	.456 (7)	.490 (9)	.404 (9)	.435 (11)	.452 (12)
bm25_issue	.409 (7)	.460 (6)	.615 (4)	.367 (14)	.418 (14)	.553 (6)

($p_{\text{Holm}} < 10^{-4}$), and for the walks alone dividing by length no longer helps at all (ppr_und_pl minus ppr_und is -0.018 on co-edited files). Under the per-file rule length-awareness is a cost everywhere, as it must be when a short file buys nothing: the length-aware fusion falls to seventh and fifth, 0.055 and 0.045 behind its unscaled twin rrf_ppr_issue_path, which becomes the best ordering under both keys.

What survives every rule is the fusion of structure with issue text: on the matched population, under each rule and on each key, a fusion is the best non-oracle ordering, and under two of the three rules a length-aware fusion is. Which structural input to fuse depends on how reading is paid for. Where a context is charged by size, as a token window is, make the walk length-aware; where it is charged by the number of files opened, do not. We state contribution 2 accordingly.

4.9 The fusion holds on repositories it was not chosen on

Both the fusion recipe and the worst-rank criterion that selects it were settled after we had seen the full results. That makes the recommendation exploratory however large its margin, so we split the corpus. A random 56 of the 112 repositories form a selection half; the remaining 56 are set aside. We apply the criterion mechanically to the selection half — among the fusions, take the one whose worst rank across the two keys is best — and then report that ordering, unexamined, on the other half. Both halves are scored on the matched population of Section 4.3, the pull requests carrying both keys, because that is the population on which a comparison across keys means anything; this is why the criterion picks rrf_pprpl_issue_path here although on the unmatched Table 10 the same criterion prefers rrf_hops_path. Splitting by repository rather than by pull request keeps every task of a repository on one side, which is what the clustering in Section 3.6 requires.

One split would not settle much: with five candidates and a winner that is already best overall, a single draw is close to a foregone conclusion. So we repeat it 200 times (Table 9). rrf_pprpl_issue_path is selected in 198 of the 200 splits, and in 195 of those it is the best non-oracle ordering on the co-edited key of the half that did not select it, 198 of 198 on the symbol key. Its mean shortfall behind the best held-out ordering is 0.002 AUC on co-edited files and 0.000 on symbol files. The two splits that select rrf_hops_issue_path instead do not confirm it.

What this establishes should not be overstated. The 200 splits are resamples of the same 112 repositories, and the selected fusion is already the best overall on the matched population, so agreement across them is largely expected: they show that the choice is *stable* — no small group of repositories is carrying it — not that it generalises to repositories we did not sample. Section 4.10

Table 9. 200 random 56/56 splits of the repositories. “Selected” counts the splits in which the worst-rank criterion picked this fusion on the selection half; “best” counts those in which it was then the best non-oracle ordering on the held-out half; “gap” is its mean AUC shortfall behind the best held-out ordering.

Method	selected	co_edited		symbol	
		best	gap	best	gap
rrf_pprpl_issue_path	198/200	195/198	.002	198/198	.000
rrf_hops_issue_path	2/200	0/2	.016	0/2	.024

Table 10. Robustness across the two evidence sources, by worst rank and by worst shortfall. Ranks are those of Table 5, oracle included, so the best attainable worst rank is 2. “Gap” is the larger of the two AUC distances to the best non-oracle ordering under each key, and does not depend on which other methods were scored. Sorted by gap; lower is better on both. The three fusions built on a length-aware structural input beat every pure signal on either measure; the two built on the unscaled walk do not.

Method	Fam.	co_edited	symbol	worst	gap
rrf_pprpl_issue_path	fus	2	6	6	.017
rrf_hops_path	fus	4	5	5	.029
rrf_hops_issue_path	fus	3	8	8	.034
rrf_ppr_issue_path	fus	5	11	11	.071
hops_lines	str	10	4	10	.097
rrf_ppr_issue	fus	9	14	14	.100
path_issue	lex	6	15	15	.108
ppr_und_pl	str	11	3	11	.114
bfs_und	str	13	10	13	.124
path_seed	lex	12	16	16	.126
ppr_und	str	14	9	14	.128
bm25_issue_pl	lex	15	17	17	.135
bm25_issue	lex	7	18	18	.150
bm25_seed	lex	8	19	19	.155
ppr_out_pl	str	17	2	17	.159
same_dir	ref	16	20	20	.171
ppr_out	str	18	7	18	.173
ppr_in_pl	str	19	21	21	.214
pagerank	glo	21	12	21	.215
indegree	glo	22	13	22	.215
ppr_in	str	20	22	22	.222
random	ref	23	23	23	.242

tests that on repositories outside the sample. Nor does it make the fusion family a discovery; it was chosen by us, from five candidates, on these data.

rrf_pprpl_issue_path is also insensitive to the one free parameter we chose without testing. Reciprocal-rank fusion’s k was fixed at the customary 60; re-running the whole pipeline at $k = 10$ and $k = 200$ moves its AUC by at most 0.007 on the co-edited key and 0.010 on the symbol key, an order of magnitude below the differences this section turns on.

Table 11. The four predictions committed to the replication package before the external repositories were scored, and what they showed. Matched population throughout except E3, mixed model with a random intercept per repository, Holm within each prediction.

Prediction	Observed	Outcome
E1 fusion in the top three non-oracle orderings under both keys, worst shortfall ≤ 0.030	rank 3 and 3 (oracle included); shortfall 0.030	pass
E2 fusion never significantly worse than four pure orderings, either key	smallest difference +0.058 ($p_{\text{Holm}} = 0.511$)	pass
E3 outward walk beats inward on the symbol key	+0.169 ($p = 0.001$)	pass
E4 names redaction costs the fusion under both keys; nothing where nothing is named	+0.030 ($p_{\text{Holm}} = 0.112$) and +0.033 ($p_{\text{Holm}} = 0.008$); control ≤ 0.003	fail

4.10 An external test on seven repositories the study never saw

Seven repositories pass every one of our sampling rules and were left out only because we restricted the task sources to SWE-bench, SWE-bench-Live and SWE-rebench: ansible, hydra, moto, modin, MONAI, pydantic and qutebrowser, whose eligible tasks come from SWE-Gym and SWE-bench Pro. We drew their tasks by the main sample's rule, seed and cap — 140 pull requests — and then, *before building a graph, a key or a score for any of them*, committed four predictions with their failure criteria to the replication package (paper/external_prereg.md); the version history records the order. The pipeline then ran unchanged. Of the 140 pull requests, 139 yield a task and 96 have a non-empty key under at least one source, 66 under both. Seven repositories are too few for the repository-level Wilcoxon test — its smallest two-sided p at $n = 7$ is 0.016 — so the predictions name the mixed model of Section 3.6 as their test, with the caveat, stated in advance, that seven groups estimate the random intercept poorly. These repositories are also larger than the sample's (a median of 289 non-test Python files against 160), and two of them are copyleft.

Table 11 gives the four outcomes. Three predictions hold. The recommended fusion is third on the oracle-inclusive scale under both keys, second only to rrf_hops_issue_path, a fusion of the same family, and its worst shortfall is 0.030 — at the limit we set, not inside it. It is never significantly worse than a pure ordering, and walking inward is as costly as in the main sample. In these larger repositories the path-matching fusions take ranks two to five under *both* keys; on the symbol key the length-aware walks no longer lead, as they do on the main sample's single-file changes.

The fourth fails. Deleting the key files' names from issues that name them costs the fusion 0.033 on the symbol key, significantly, but 0.030 on the co-edited key with $p_{\text{Holm}} = 0.11$, on only 33 matched pull requests in six repositories; the control part — the redaction moves nothing where nothing is named — holds. So the causal effect of the names alone is not established on the co-edited key here, and where it is measured it is half the size it is in the main sample. The wider arms, which we did not predict, cost the fusion 0.086 and 0.090 on the co-edited key: in these repositories an issue points at its files through their directories far more than through their names, which is what one would expect of code bases whose modules are named for deep package paths. The share of issues naming a key file as code, 40.6%, is the main sample's to within half a point.

4.11 A human-annotated key

Both of our keys are proxies, and the symbol key is built along the same import edges the structural orderings walk. ContextBench [12] annotates by hand the code an agent must see to reconstruct a passing fix. Its 409 Python tasks from SWE-bench Verified and SWE-bench Pro come from datasets we already hold, so we ran them through the unchanged pipeline with a third key, gold: the

Table 12. The four predictions committed before the ContextBench tasks were scored, and what the human-annotated key showed. G1–G3 on the pull requests with a non-empty gold key, mixed model with a random intercept per repository, Holm within each prediction.

Prediction	Observed	Outcome
G1 fusion in the top three non-oracle orderings on the human key, shortfall ≤ 0.030	rank 3 (oracle included); shortfall 0.009	pass
G2 fusion never significantly worse than four pure orderings	smallest difference +0.086 ($p_{\text{Holm}} < 0.001$)	pass
G3 outward walk beats inward	+0.083 ($p = 0.006$)	pass
G4 Kendall $\tau \geq 0.5$ between the human key's ranking and each proxy's	symbol 0.59, co-edited 0.97	pass

Python files in a task's annotated context, less the seed, restricted to graph nodes like the others. We drew the tasks by the main sample's eligibility rule, seed and cap — 203 pull requests in 15 repositories, ten of which are in the main sample, as are 24 of the pull requests — and committed four predictions before building any graph or score (paper/contextbench_prereg.md). This is a test of the keys, not of generalisation to new code. Of the resulting tasks, 215 from 108 pull requests in 14 repositories have a non-empty gold key. A gold key has a median of 3 files; 59% of its files are also co-edited files and 63% are also symbol files.

All four predictions hold (Table 12). The fusion is the second-best non-oracle ordering on the human key, 0.009 behind `rrf_hops_issue_path`; it beats each of the four pure orderings we named by 0.086 to 0.128 ($p_{\text{Holm}} \leq 0.008$); walking outward beats walking inward by 0.083; and the ranking of all twenty-two orderings under each proxy key agrees with the ranking under the human key (Kendall $\tau = 0.59$ for the symbol key, 0.97 for the co-edited key).

The more informative result is one we did not predict: *the human key sides with the co-edited key* (Table 13). Under it the lexical and fused orderings rank as they do under the co-edited key, while the length-aware walks, second and fifth under the symbol key on these same tasks, fall to eleventh and fifteenth. The symbol key's preference for structure is not corroborated by people's judgement of what had to be read. That is what the circularity of its construction predicts, and it means the structural wins on the symbol key in this paper are statements about that key. An objection is that the gold key contains the edited files and agrees with the co-edited key by construction. So we also scored, as an exploratory analysis added after the predictions were tested, the gold files that the fix *reads but does not edit* — the context a graph walk is supposed to be best at finding — on the 85 pull requests that have any. There the walks and the issue-only retrievers are indistinguishable (differences of 0.006 to 0.012, $p_{\text{Holm}} = 1$), and the fusion leads: it beats the issue-only orderings by 0.082 and 0.084 ($p_{\text{Holm}} \leq 0.008$) and the walks by 0.074 to 0.090, which at this size does not survive correction ($p_{\text{Holm}} \geq 0.12$).

The set is small — 108 pull requests in 14 repositories — and its annotators saw the fix, so their notion of necessary context is itself a reconstruction. It is nonetheless the only key in this paper that no part of our pipeline built, and the recommendation holds on it.

4.12 RQ6: what a budget rules out, and what it does not

A coverage number says how much of the key a method found. It does not say whether finding all of it was possible. If the needed files together exceed the budget, no ordering of them fits and no downstream system can succeed, however capable: a model cannot use a file it was never given. That bound is a property of the task and the budget alone, so it can be measured once and applied to every method at once — and, usefully here, without running a model.

Table 13. AUC (and oracle-inclusive rank among 23) on the ContextBench tasks under the human key, its read-only part (gold files the fix does not edit; exploratory), and the two proxy keys.

Ordering	gold	read-only	co_edited	symbol
rrf_hops_issue_path	.379 (2)	.347 (2)	.436 (2)	.343 (4)
rrf_pprpl_issue_path	.370 (3)	.341 (3)	.423 (3)	.349 (3)
rrf_hops_path	.345 (5)	.317 (5)	.387 (4)	.338 (6)
bm25_issue	.283 (7)	.259 (10)	.333 (6)	.217 (17)
path_issue	.280 (9)	.248 (13)	.322 (7)	.228 (15)
hops_lines	.275 (10)	.274 (7)	.281 (10)	.309 (7)
ppr_und_pl	.261 (11)	.267 (8)	.261 (11)	.352 (2)
ppr_out_pl	.230 (15)	.249 (12)	.202 (16)	.339 (5)
pagerank	.180 (18)	.192 (18)	.159 (20)	.261 (12)
random	.103 (23)	.104 (23)	.104 (23)	.095 (23)
oracle	.694	.699	.739	.731

Table 14. Share of pull requests whose *entire* key fits within a token budget. This is a ceiling: no ordering can exceed it, and no agent can act on files outside it.

Token budget	co_edited (%)	symbol (%)
4,000	34.1	31.5
8,000	54.2	50.8
16,000	75.1	72.5
32,000	90.3	87.3
64,000	96.9	95.5
128,000	99.5	99.4
200,000	99.9	99.9
1,000,000	100.0	100.0

Table 14 reports it. The complete answer fits a 32,000-token budget for 90.3% of pull requests under the co-edited key and 87.3% under the symbol key, and a 64,000-token budget for over 95% of both. The task is not, in general, impossible at realistic budgets.

Table 15 applies the same all-or-nothing criterion to the orderings, in the frozen unit of lines: the share of pull requests each one takes to *complete* coverage within 8,000 lines. The oracle succeeds on 97.5%. The best ordering on the co-edited key, rrf_hops_issue_path, completes 58.5% of pull requests, and the best on the symbol key, hops_lines, 51.0%.

The oracle reads only key files, so its success is partly definitional: fitting a small key into a large budget is close to arithmetic. We therefore do not read the forty-point gap as headroom that better ordering could be expected to recover; it is an upper bound on what better selection could add, not an estimate. What the comparison does establish is the converse. At 8,000 lines the budget is almost never the binding constraint, so an ordering that fails to cover the key fails because of the order, not the budget — which rules out the explanation a reader is likely to reach for first.

Two medians in this paper look as though they disagree and do not. The median *task* needs 1,150 lines read before the co-edited key is fully covered (Table 6), while Table 14 says 54.2% of *pull requests* fit inside 8,000 tokens. Averaging a pull request's tasks first gives a median of 991 lines, and at the median 7.88 tokens per line that is about 7,800 tokens — just inside the budget, which is why just over half of them fit.

Table 15. Share of pull requests taken to *complete* coverage within 8,000 lines. The oracle row is the ceiling of Table 14 expressed in the frozen unit. The reversal of Table 5 reappears under this stricter criterion.

Ordering	co_edited (%)	symbol (%)
oracle	97.5	95.9
rrf_hops_issue_path	58.5	49.3
rrf_pprpl_issue_path	56.5	49.1
rrf_hops_path	54.5	50.3
rrf_ppr_issue_path	53.4	45.0
bm25_issue	51.8	36.9
rrf_ppr_issue	50.6	44.6
bm25_seed	48.8	36.6
path_issue	48.4	36.9
hops_lines	48.1	51.0
ppr_und_pl	45.1	50.0
ppr_und	45.0	50.1
bm25_issue_pl	39.8	33.9
path_seed	39.6	36.3
bfs_und	36.6	40.1
ppr_out_pl	35.4	50.1
ppr_out	34.5	49.7
ppr_in	34.0	22.8
ppr_in_pl	33.9	22.3
pagerank	31.9	43.4
indegree	28.2	38.9
same_dir	27.8	24.2
random	15.3	14.3

The reversal of Table 5 survives this stricter criterion: `bm25_issue` loses 14.9 points and `path_issue` 11.5 points moving from the co-edited key to the symbol key, while the outward walks gain the most (`ppr_out` 15.2 points, `ppr_out_pl` 14.7) and `pagerank` gains 11.5. The gain is not a property of the structural family as such: the two inward walks lose 11.2 and 11.7 points, about as much as `path_issue`'s 11.5 and in the same direction. It is a property of walking the direction the key was built along, not of using the graph. And the fusion orderings again sit at or near the top of both columns, which is the same conclusion RQ5 reached by AUC, reached here by a measure that ignores partial credit entirely.

4.13 Do the lexical methods just copy paths out of the issue?

Issue reports often contain stack traces, and a stack trace names files. If the lexical advantage came from reports quoting the paths of the files that had to change, a benchmark average would describe issues that name their files, not issues in general — which is the sense in which we call it leakage. For a deployed tool the names are a legitimate input.

Much of the advantage does come from such issues, and the effect is big enough that we report it as a result rather than as a reassurance.

The full path of a key file appears verbatim in the issue text for 12.6% of tasks, and a bare file stem for 48.5%. Path BM25 tokenises a path into lowercased subtokens, so a stem is enough to match and the full-path control is far too weak. But a bare stem is also too strong a criterion on its own: an issue about “models” and a key file called `models.py` share a word because a project's prose

and its filenames draw on one domain vocabulary, and using that overlap is retrieval, not leakage. We therefore classify every stem occurrence by how it is written. A mention is *explicit* if it appears as code — `core.py`, `simbad/core`, a dotted module path, inside backticks, or inside a fenced or indented block — and *incidental* if the stem occurs only as an ordinary English word. Explicit mentions cover 40.2% of tasks. A further 8.3% have incidental matches only, the two together making up the 48.5% of tasks with any stem at all.

That gives three nested strata: issues quoting no full key path (2,108 tasks), issues naming no key file as code (1,442 tasks), and the strictest set containing no key stem at all (1,243 tasks). We treat the middle one as primary. It removes the leakage channel while leaving the vocabulary signal in place, and it does not select as heavily for vague issues as the strictest does.

Table 16 applies this control *together with* the population control of Section 4.3, recomputing within each stratum the set of pull requests that carry both keys. Applied on its own, the leakage control would leave the symbol columns full of the single-file pull requests that the matching control exists to remove, and a claim that survives each control separately need not survive both. Reading the strata jointly is the only version of this table we regard as evidence.

On co-edited files, path matching falls and structure does not follow it down. On the full corpus `path_issue` is sixth of the twenty-three orderings. Under the weak full-path control it barely moves, falling from 0.434 to 0.408 and holding sixth. Under the primary control it falls to 0.363 and *thirteenth*; under the strictest, to 0.339 and *fourteenth*. `hops_lines` — pure structure, no issue text — rises from tenth to sixth in both. Because the middle and the strictest strata agree, the finding is not an artefact of over-correcting: it does not depend on discarding incidental vocabulary overlap.

The direction of that change is clear; its significance is weaker than the ranks suggest, and we report the tests rather than the gap. `hops_lines` over `path_issue` on co-edited files is -0.007 under the full-path control ($p_{\text{Holm}} = 1$), $+0.042$ under the primary control ($p_{\text{Holm}} = 0.60$) and $+0.056$ under the strictest ($p_{\text{Holm}} = 0.04$). Only in the strictest stratum does pure structure beat path matching outright. In the primary stratum the two become *indistinguishable*, which is already the interesting claim: the ordering that appeared to be the best single signal on this key stops being distinguishable from one that never reads the issue, once the issues that name the answer are removed. These strata are smaller than the corpus — 531 pull requests across 109 repositories in the primary stratum — so the failure to separate them is in part a matter of power, and we do not read it as evidence of equivalence.

On symbol files, length-aware walks lead in the two stricter strata. Under the full-path control the fusion still leads (0.456, against 0.440 for `ppr_und_p1`). Under the primary control and the strictest, `ppr_und_p1` and `hops_lines` lead instead, second and third on the oracle-inclusive scale in the primary stratum, at 0.445 and 0.435, with the best fusion fourth at 0.430. The gap is not significant ($+0.017$ for `ppr_und_p1`, $p_{\text{Holm}} = 1$), so this is a place where the fusion stops leading rather than a place where it loses.

The seed is not the explanation. The `path_seed` control separates “issue text” from “path matching” directly: the same path-BM25 ranking, queried by the seed’s own path instead of the issue, scores 0.371 on co-edited files against 0.434 for the issue query ($+0.058$, $p_{\text{Holm}} < 10^{-4}$, repository as the unit), and the two are indistinguishable on symbol files ($+0.020$, $p_{\text{Holm}} = 0.81$). For comparison, the issue names the seed file — the file every structural method is handed for free — less often than it names a key file: its full path appears in 10.3% of tasks and its stem in 35.1%, against 12.6% and 48.5% for the key files.

Table 16. Both controls at once: AUC and rank (in brackets) under the three nested leakage strata, each recomputed on the pull requests within that stratum that carry both keys. Ranks are over all twenty-three orderings with the oracle included, the scale of Table 5.

Method	no full path		no explicit mention		no path or stem	
	co_edited	symbol	co_edited	symbol	co_edited	symbol
rrf_pprpl_issue_path	.468 (2)	.456 (2)	.434 (2)	.430 (4)	.417 (3)	.414 (5)
rrf_hops_path	.441 (4)	.445 (3)	.410 (4)	.421 (6)	.392 (4)	.404 (7)
ppr_out_pl	.325 (17)	.419 (7)	.335 (16)	.427 (5)	.331 (16)	.423 (4)
ppr_und_pl	.374 (11)	.440 (5)	.378 (10)	.445 (2)	.373 (10)	.441 (2)
hops_lines	.390 (9)	.427 (6)	.394 (6)	.435 (3)	.387 (6)	.429 (3)
path_issue	.408 (6)	.383 (11)	.363 (13)	.343 (19)	.339 (14)	.320 (20)
bm25_issue	.397 (7)	.357 (15)	.384 (7)	.346 (16)	.385 (9)	.346 (17)
pagerank	.274 (21)	.356 (16)	.281 (21)	.360 (14)	.285 (21)	.361 (13)
<i>pull requests</i>	655	655	531	531	487	487

4.14 RQ7: what does reading the issue actually add?

The previous subsection removes tasks. That answers whether a ranking survives the removal, but it cannot say what the issue text contributes, for two reasons. Every ordering is affected by which tasks are dropped; and the tasks that are dropped are not a random sample of issues. A report containing a stack trace is plausibly longer, more technical and more tractable than a one-line feature request, so a comparison across leakage strata compares kinds of issue as well as degrees of leakage. We answer the question twice, once in each direction, and the two answers together decompose the effect.

Across tasks: an issue-free twin. For each of our two leading fusions we construct a twin in which every issue-derived input is replaced by its seed-derived counterpart. `rrf_pprpl_issue_path`, which fuses a length-aware walk with BM25 over file contents and BM25 over file paths, both queried by the issue, is paired with `rrf_pprpl_seedpath`, which fuses the same walk with the same two BM25 rankings queried by the *seed file's own path and contents*. The twin sees the same graph, blends the same number of lists through the same rank fusion, and runs the same retrieval machinery. It simply never reads the issue. The difference is what the issue is worth, net of the “a fusion just gets more input” explanation the fusion result otherwise invites.

Over the corpus the issue is worth +0.069 AUC on the co-edited key and +0.055 on the symbol key, both strongly significant (Table 17). Split by the leakage classification, that becomes +0.156 and +0.121 on the 40.2% of tasks whose issue names a key file as code, against +0.016 and +0.012 on the rest, where neither survives correction; requiring the stem to be absent as well takes it to +0.003 and −0.002. The progression is monotone across five strata and holds for both pairs and both keys.

Because these are null results we report what they exclude rather than treating them as zeros. The bootstrap interval on the no-mention stratum is [0.000, 0.031] on the co-edited key and [−0.003, 0.027] on the symbol key, so the data are consistent with the issue being worth up to about 0.03 AUC there, and inconsistent with its being worth anything close to the 0.156 it is worth where it names a file. Taken alone this comparison would say that the issue is worth almost exactly what it names — and taken alone it would be overstating, because it is confounded with which issues, and which tasks, name their files.

Within task: redaction. So we did the other experiment. For every task we take the *same* issue and delete only the occurrences of its key files' names that our classifier calls explicit, leaving the prose, the traceback, the surrounding directory names and the length untouched. Scoring the same task on both versions isolates the names themselves: the issue, the writer, the project and the difficulty are all held fixed.

What counts as "the names" is a choice, so we make it three times. The *names* arm removes only the stem and any immediately following `.py`: `astroquery/simbad/core.py` becomes `astroquery/simbad/`. That leaves a channel open, since path BM25 tokenises paths and the surviving directory names still point at the file, so the *paths* arm also deletes every component of each key file's path wherever it is written as code. Neither touches a traceback's in `query_object` or a backticked `Simbad.query_object`, which a content ranking can match against the file that defines it, so the *symbols* arm also deletes every function, class and method name defined in a key file, again only where written as code. The names arm is a lower bound on leakage. The wider arms over-remove — a directory called `models` or a method the user calls by name is also ordinary vocabulary — so the three bracket the effect rather than measure it.

The result is in Table 18. Of the 970 tasks whose issues name a key file, the 848 in the matched population — 378 pull requests in 105 repositories — are scored; on them, removing just the names costs `path_issue` 0.110 AUC on the co-edited key and 0.092 on the symbol key, and the best fusion 0.061 and 0.050. Every one of these survives Holm correction with the repository as the unit, and the bootstrap intervals are narrow: for `path_issue` on co-edited files it is [0.090, 0.131], and the widest of the eight spans 0.041. The wider arms cost more, as they must: 0.148 and 0.157 for `path_issue` on co-edited files, 0.084 and 0.096 for the fusion. On the 1,072 matched tasks (531 pull requests) among the 1,442 where the names arm found nothing to remove, it changes nothing — at most 0.002 in every cell, significant in none, which is the sanity check the design demands. The wider arms do find something there, and cost the fusion 0.016 to 0.018: an issue that never names a key file can still quote its directory or a function it defines.

The decomposition. Both experiments run on the same matched population with the repository as the unit, so their answers add up, and together they say something neither says alone. On the tasks whose issue names a key file, the issue as a whole is worth +0.156 AUC to the best fusion on the co-edited key and +0.121 on the symbol key. Of that, the names account for +0.061 and +0.050, and the rest of the issue for +0.095 and +0.071, every term significant under Holm correction. On the tasks whose issue names nothing, the issue as a whole is worth +0.016 and +0.012, and neither is significant.

So both effects are real and neither subsumes the other. Filename leakage is causal and large: path matching loses a fifth of its score on the affected tasks when only the names go, and nearly three tenths under the widest arm. A reader of the stratified table alone would put the whole +0.140 stratum difference down to leakage. Under the names arm +0.061 of it is leakage and +0.079 is not; under the widest arm, taking into account that it also removes something from issues that name nothing, the split is +0.078 and +0.062 (on the symbol key, +0.050 to +0.065 of a +0.109 gap). What remains after every redaction we can define is not leakage we can find, and we do not claim it is issue *quality* either: it is a difference between the issues that name files and the ones that do not, and between the tasks they describe, that the redaction cannot reach. Even `path_issue`, which reads nothing but file paths, still scores 0.393 on the named-file tasks after the widest redaction against 0.333 on the others after the same treatment, so some of that difference is in what the issues say or in which files the tasks need, not in how the issues are written.

What this licenses. It does not show that issue text is useless for code retrieval; developers use it, and an ordering that exploits a real report naming a real file exploits something that exists. What

Table 17. What the issue is worth: each fusion minus its issue-free twin, in AUC, with the repository as the unit and the population matched to carry both keys. Positive means the issue-aware ordering is better. The first row is not a control but the complement of the fourth: it is the tasks whose issue *does* name a key file as code. Starred differences survive Holm correction at 0.05.

Issue stratum	repos	rrf_pprpl_issue_path		rrf_hops_path	
		co_edited	symbol	co_edited	symbol
names a key file as code	105	+0.156*	+0.121*	+0.148*	+0.114*
every task	111	+0.069*	+0.055*	+0.053*	+0.038*
no full path quoted	111	+0.047*	+0.036*	+0.033*	+0.022*
no explicit mention	109	+0.016	+0.012	−0.001	−0.009
no stem at all	107	+0.003	−0.002	−0.008	−0.018

Table 18. Redaction, within task: AUC on the issue as written, and the change when the same issue is redacted by each of three arms — the key files’ names only; the names and every component of their paths; and those plus every function, class and method name defined in a key file, each deleted only where written as code. The pull requests are those of the matched population inside each leakage stratum, with the repository as the unit; Holm-corrected within each arm, and a star marks $p < 0.05$. The names arm must change nothing in the lower block, and does not; the wider arms can, because an issue that names no key file can still quote its directory or a function it defines.

Method	co_edited				symbol			
	AUC	names	paths	symbols	AUC	names	paths	symbols
<i>issue names a key file as code (378 pull requests in 105 repositories)</i>								
path_issue	.550	−0.110*	−0.148*	−0.157*	.498	−0.092*	−0.125*	−0.134*
bm25_issue	.452	−0.009*	−0.011*	−0.026*	.408	−0.006*	−0.008*	−0.021*
rrf_hops_path	.529	−0.072*	−0.097*	−0.105*	.515	−0.059*	−0.082*	−0.089*
rrf_pprpl_issue_path	.566	−0.061*	−0.084*	−0.096*	.537	−0.050*	−0.074*	−0.082*
<i>issue names no key file as code (531 pull requests in 109 repositories)</i>								
path_issue	.363	−0.000	−0.025*	−0.030*	.347	−0.002	−0.026*	−0.028*
bm25_issue	.376	−0.000	−0.003*	−0.006*	.341	−0.000	−0.002*	−0.004*
rrf_hops_path	.415	+0.001	−0.017*	−0.019*	.426	−0.001	−0.020*	−0.019*
rrf_pprpl_issue_path	.434	+0.002	−0.016*	−0.018*	.433	+0.000	−0.018*	−0.017*

it shows is that a benchmark assembled from issue reports of this kind measures two things at once, and that the two can be bracketed for the cost of a few regular expressions. Any evaluation on SWE-bench-derived data that reports a gain from issue-aware retrieval and does not stratify is reporting their sum. That, rather than the ranking of our own methods, is the finding we would most like the field to take up.

4.15 Without a free seed

Every task so far hands the orderings the file being changed. An agent is not handed it: it must find that file from the issue, and localisation is where agents are known to struggle. To see whether the recommendation depends on the free seed, we removed it. For each of the 1,162 pull requests the issue alone picks a starting file — the top of the reciprocal-rank fusion of BM25 over contents and BM25 over paths, the two issue-only rankings used throughout — and every seed-conditioned

Table 19. Without a free seed: mean AUC per pull request when the seed is the issue’s own best guess. *Edited*: every edited source file, the true seeds included. Each key is split by whether the issue names an edited file as code.

Ordering	edited			symbol		
	all	named	not	all	named	not
rrf_pprpl_issue_path	.480	.533	.427	.440	.478	.403
rrf_hops_path	.454	.515	.393	.424	.470	.378
<i>issue-only fusion (picks the seed)</i>	.471	.524	.417	.396	.436	.356
path_issue	.458	.531	.383	.397	.451	.342
bm25_issue	.417	.429	.405	.355	.369	.341
ppr_und_pl	.404	.448	.361	.429	.456	.402
hops_lines	.404	.446	.362	.404	.433	.375
random	.241	.241	.242	.242	.244	.240
oracle	.758	.763	.753	.760	.762	.758

ordering walks from that guess and reads it first. Issue-only orderings need no seed and run unchanged over the whole repository. The unit is the pull request, and the two keys become every existing source file it edits, the true seeds included, so that finding the file to change is part of what is scored; and every file defining a name the change uses. This analysis was added in response to review and is exploratory.

The guess is right in 41% of pull requests: in 53% of the 583 whose issue names an edited file as code, and in 29% of the 579 whose issue does not. (Half the pull requests, not two fifths of the tasks, count as naming here, because the seed’s own name now counts.)

Table 19 gives the result. Walking from the guess and fusing with the issue remains the best non-oracle ordering under both keys, at 0.480 and 0.440, and in no stratum is it significantly worse than any alternative. What the walk adds depends on the key. On the symbol key it adds 0.044 to the issue-only fusion that chose its seed ($p_{\text{Holm}} < 10^{-4}$), and does so whether or not the issue names a file: even a wrong guess is usually in the right package, and the walk reaches what the change depends on from there. On the edited files it adds 0.009 ($p_{\text{Holm}} = 0.03$) and nothing significant within either stratum: when the task is to find the files to change, what the issue retrieves directly is nearly all there is. The walks on their own, without the issue, fall well behind on the edited files (0.404 against 0.480), because the file they start from is wrong more often than right.

This is the simplest possible localiser, and an agent’s is better; the numbers bound what a graph adds on top of a weak guess, not a strong one. But the recommendation does not depend on the free seed.

4.16 Sensitivity: does the uncontrolled ranking survive in large repositories?

The corpus skews small — the median repository contributes 131 graph nodes — so the reversal has to be shown not to be an artefact of repositories one can almost read whole. Table 20 repeats the ranking within quartiles of repository size (quartile boundaries 73, 131 and 246 nodes; the largest repository has 2,487). The median of 131 graph nodes and the median of 286,385 tokens per repository are consistent: they imply roughly 2,200 tokens, or some 260 lines, per file, which is an ordinary Python module.

The pattern of Table 5 holds in every quartile, so the uncontrolled ranking is not an artefact of repositories one can almost read whole. This analysis runs on the full corpus without either control, so it says nothing about whether the qualifications of Sections 4.3 and 4.13 are size-dependent; we report it as a robustness check on Table 5 alone. The gap does not widen monotonically with size:

Table 20. Mean AUC (rank over all twenty-three orderings, oracle included) by quartile of repository size in graph nodes. $Q1 \leq 73$, $Q2 \leq 131$, $Q3 \leq 246$, $Q4$ up to 2,487.

Method	co_edited				symbol			
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
rrf_pprpl_issue_path	.574 (2)	.487 (2)	.418 (2)	.369 (2)	.568 (3)	.476 (2)	.403 (5)	.375 (3)
rrf_hops_path	.552 (4)	.465 (4)	.397 (4)	.330 (5)	.562 (6)	.473 (3)	.409 (4)	.362 (5)
ppr_out_pl	.473 (16)	.359 (17)	.268 (16)	.168 (17)	.562 (5)	.450 (7)	.414 (3)	.372 (4)
ppr_und_pl	.497 (13)	.399 (9)	.316 (11)	.241 (12)	.568 (4)	.467 (4)	.423 (2)	.387 (2)
hops_lines	.517 (7)	.412 (7)	.329 (9)	.244 (11)	.569 (2)	.454 (6)	.402 (6)	.343 (8)
path_issue	.537 (6)	.437 (5)	.372 (6)	.302 (6)	.515 (14)	.416 (10)	.324 (13)	.257 (15)
bm25_issue	.516 (8)	.402 (8)	.334 (7)	.287 (7)	.476 (18)	.367 (18)	.273 (18)	.233 (17)
pagerank	.442 (21)	.299 (22)	.199 (21)	.113 (22)	.521 (11)	.370 (17)	.333 (12)	.286 (13)

the co-edited-minus-symbol rank gap for ppr_und_pl runs 9, 5, 9, 10 across the quartiles and for hops_lines 5, 1, 3, 3, so the right claim is stability, not a trend. In the largest quartile, issue-text path matching is sixth on co-edited files and fifteenth on symbol files, while length-aware personalised PageRank is twelfth and second respectively. The query-independent prior degrades fastest of all: on co-edited files in the largest quartile it scores 0.113 against a random floor of 0.087, which is to say that in the repositories where retrieval matters most, knowing only which files are central is worth almost nothing. rrf_hops_path never falls below sixth in any of the eight cells, and is fourth or better in five of them.

5 DISCUSSION

What we can and cannot say about the two keys. The tidy version of this paper would be that the import graph answers the compiler's question and the issue text answers the developer's, and that the field has been grading the first against the second. Our own controls will not support the tidy version.

On the symbol key, structural methods win on the full corpus and are indistinguishable from issue-to-path matching once the population is matched to the co-edited key (Section 4.3). That key is also built by following imports, so we would have discounted a win there in any case. On the co-edited key, issue-to-path matching wins on the full corpus and becomes indistinguishable from pure structure once tasks whose issues name a key file in code are removed, losing to it only in the strictest stratum (Section 4.13). Each half of the reversal, in other words, has an explanation that is not about the signals at all — one is a population effect, the other a leakage effect. The uncontrolled comparison reports both halves as findings about the signals; the controlled comparison does not.

Calling the first of these an artefact would undersell it. Single-file pull requests are 454 of the 1,156 our symbol key covers, and they are real work, not a sampling accident. What the matching control shows is that structure's advantage on the symbol key is *concentrated* in them: on those 454 the best directed walk scores 0.605 and the five best non-oracle orderings are all structural, against 0.417 for the same walk on the matched population. That is a finding in its own right, and an intuitive one — when a change touches one file, what it needs is most likely to be what that file imports. It is not evidence that structure beats words in general, which is what the unmatched table appears to say.

A human-annotated key lets us say which proxy to believe where they disagree, and it sides with the co-edited key (Section 4.11). On ContextBench's tasks the length-aware walks drop from second and fifth under the symbol key to eleventh and fifteenth under the human one, and on the context a fix reads without editing — the files a graph walk is meant to surface — walks and issue

text are indistinguishable. The symbol key’s preference for structure is, on this evidence, largely a property of how that key is built.

What remains is weaker but real. The two keys still induce different rankings, and a method tuned on one is not thereby good on the other. But the summary of the single-signal comparison is that neither family dominates once both confounds are controlled, and the interesting result is elsewhere.

Two effects, not one. The result we would most like carried forward is Section 4.14, and its interest is that it separates two things the literature reports as one number: filename leakage, which is causal and large, and a residual difference between the issues that name files and those that do not, which no redaction we could define removes. Both halves matter here. Ignore the first and a lexical baseline looks stronger than it is; ignore the second and the stratified comparison looks like proof that issue text does nothing, which our own within-task experiment contradicts. We resist the second reading in particular: the null in the no-mention stratum bounds the issue’s contribution at about 0.03 AUC on 531 pull requests. It does not show it is zero.

Fusion is the safe choice. Reciprocal-rank fusion of a length-aware walk with issue-to-path matching is the best non-oracle ordering under both keys on the matched population, where it beats the best pure lexical ordering by 0.055 and 0.067 and the best pure structural one by 0.097 and 0.039 at the repository level. Across 200 random repository splits a mechanical criterion selects it on the choosing half in 198, and in 195 of those it is the best non-oracle ordering on the half that did not choose it (Section 4.9): the choice is stable across the repositories we have, which is not the same as having been confirmed on repositories we do not.

The claim is robustness rather than dominance. Under the joint control of Table 16 the fusion is first or second on co-edited files in every stratum, behind only another fusion where it is not first, but sits behind two length-aware walks on symbol files in the two stricter strata, by margins that do not survive correction ($p_{\text{Holm}} = 1$). On the matched population, in every leakage stratum and on either key, the fusion is never significantly worse than the best pure ordering and is often significantly better. On the full symbol-key population it is not: once the tasks whose issue names a key file are removed, the length-aware walks beat it by 0.049 to 0.062 ($p_{\text{Holm}} < 10^{-4}$), because that population keeps the 454 single-file pull requests in which structure’s advantage is concentrated. For a change that touches one file, the walk alone is the better choice. That is what we would recommend on, and it is a weaker claim than “fusion wins”.

Three checks added in response to review bear on how far it travels. It is not an artefact of the cost rule: under a rule that skips an oversized file instead of stopping, the fusion remains best or within 0.003 of best, though the part of its margin owed to length-awareness shrinks by about sixty per cent, and under a per-file rule the unscaled fusion is the one to use (Section 4.8). It is not an artefact of our keys: against a human-annotated key it is second of twenty-two, 0.009 behind a fusion of the same family. And it does not depend on being handed the file to change: when the issue’s own best guess is used as the seed, right in only 41% of pull requests, walking from the guess and fusing still gives the best non-oracle ordering under both keys (Section 4.15).

We were also able to test the obvious alternative explanation. The fusions are the only methods given both the seed and the issue, so “a fusion just gets more input” is the null for the whole result. The issue-free twins of Section 4.14 answer it: blending a second list does help a little by itself — `rrf_pprpl_seedpath` beats its own walk `ppr_und_pl` by 0.049 on co-edited files — but the issue-aware fusion beats the twin by a further 0.065, both on the full corpus with the repository as the unit. Complementarity is real, and between a third and seven tenths of what the issue adds to it is leakage: on the matched population the issue is worth 0.069 to the fusion over all tasks, and the three redaction arms cost 0.023, 0.043 and 0.048 of it.

The field's default key, restated carefully. Every repository-context system in Table 1, graph-based or not, is graded against the files the gold patch touched — our `co_edited` source. On that source, and given a correct seed file, the best purely structural ordering trails matching issue words against file paths by 0.039 AUC over the full corpus ($p_{\text{Holm}} = 0.07$ with the repository as the unit; $< 10^{-3}$ in the mixed model); once the issues naming a key file are removed the two are indistinguishable. What matters for those evaluations is that the comparison depends on a property of the issue corpus that none of them reports for its own evaluation.

Two further qualifications matter. Our main task hands every structural method a correct seed file for free, whereas LocAgent, CoSIL and the rest must localize from the issue text alone. Section 4.15 removes the free seed in the simplest way — the top file of an issue-only lexical ranking becomes the seed — and finds that the walk still adds to what the issue alone retrieves, clearly on the symbol key and marginally on the edited files, but a learned localiser inside an agent loop is a different component, and its graph may earn its place differently. And they are LLM agents that also read the issue, so the graph is one component among several. This paper is therefore not evidence about how well their graph components perform in their own setting. What it shows is that, given a correct seed or the issue's own guess at one, graph proximity is not the strongest single signal on the key those systems adopt, and that the comparison depends on how often the issues name the files being scored.

Naming the answer deserves to be a standard stratum. Two tasks in five in this corpus have an issue that names a key file as code. That is not a benchmark defect — real reports really do contain stack traces — but it means any lexical baseline evaluated without the stratification is reporting a mixture of two very different regimes, and any structural method compared against such a baseline is being judged against a moving target. The control costs a regular expression over text the benchmark already ships. We think retrieval papers should report the rate and the stratified result as a matter of course, in the same way they report the corpus size.

What “within two hops” is worth. The two-hop neighbourhood earns its reputation on recall and loses it on precision: it contains seven of every eight needed files and also half the repository. Reporting the recall without the size is, on this corpus, reporting half a statistic.

What this means for an agent's context, and what it does not. The budget framing is not decoration: at a median 286,385 tokens, most of these repositories do not fit in the windows current systems run with, so selection is forced rather than optional. Two numbers bound what selection can achieve. The answer is a median 2.98% of the repository, so a well-chosen context could be some 34× smaller than the whole; and an oracle ordering fits the complete answer inside 8,000 lines for 97.5% of pull requests where our best method manages 58.5%.

We do not show that better selection raises repair rates. We did not run an agent, and a retrieval improvement need not survive contact with one. What the ceiling licenses is the weaker statement: a file absent from the context cannot be used, so the share of tasks whose answer fits the budget bounds any system operating under it. It does not license the claim that the remaining gap is reachable from the signals we have, only that it is not closed off by the tasks themselves. Section 6 specifies the experiment that would convert the bound into a performance claim, in advance of running it.

A note on order versus selection. An ordering plus a budget is a selection, and it is the selection that matters here. We make no claim about the position of a file *within* a prompt, which is a different question with its own literature; every result in this paper concerns which files are read at all.

The human reading of the same result. The budget for a developer is attention rather than tokens, and the results have a plain reading there. Asked to fix the login page, a developer thinks of the authentication files partly because the words match and partly because the imports connect; our fusion result says those two routes are largely complementary, and our leakage result says that how much the words help depends on how explicitly the report was written. We report the token framing because it makes the budget concrete and measurable; we do not think the finding is confined to machines.

Recommendations. (1) Report retrieval results separately for at least one structural and one behavioural notion of relevance, and report the worst case across them. (2) Report, and stratify by, whether the query text names the answer files in code. (3) Match populations before comparing across keys; a key that only exists for multi-file changes does not cover the same tasks as one that does not. (4) Treat a query-independent repository prior, not random order, as the floor for structural methods — on symbol-style keys it reaches 83% of a personalised walk’s absolute AUC, or 67% of the walk’s margin over a random ordering, which is the fairer reading. (5) Where a graph-based tool orders context by structure alone, add a lexical signal from the issue text and make the structural side length-aware before fusing; where one already uses both, report them separately as well as combined. (6) Report cost in tokens and report the ceiling alongside the score, so a reader can tell a hard task from a poor ordering. (7) Cluster paired tests by repository; pull requests within a repository are not independent, and ignoring that overstates significance — here 102 of 1,076 comparisons significant by pull request are not significant by repository. (8) Before attributing a gain to issue text, run the two controls of Section 4.14: delete the answer files’ names from the issue and re-score the same tasks, and compare against the same method with the issue replaced by a seed-derived query. Redact more than the filenames: on our corpus deleting the names alone costs the best fusion 0.061 AUC, and deleting their directories and defined symbols too costs 0.096, so a names-only redaction is a lower bound.

6 A PROTOCOL FOR THE DOWNSTREAM TEST, FIXED IN ADVANCE

This paper measures retrieval, not repair. The obvious next question — whether a context assembled by one of these orderings lets a model actually fix the issue — requires running models, which we have not done. Rather than gesture at it as future work, we specify it here in enough detail to be executed or contradicted, in the same spirit as the design freeze that preceded this study (§3). We commit to reporting the result whichever way it falls.

Corpus. The 1,162 scored pull requests of this study. Every one carries a test specification from the benchmark suite it came from, naming the tests that must change from failing to passing and those that must keep passing, so resolution can be decided by execution rather than by patch similarity. We note that only 15 of them overlap SWE-bench Verified, so this is not a re-run of an existing leaderboard.

Conditions. Eight contexts per task, each truncated to the same token budget. A repair experiment has no evidence key — the model gets one context and either the tests pass or they do not — so every condition names one concrete ordering: (i) the seed file alone, a floor; (ii) seed plus ppr_und_pl, the structural input of the recommended fusion (iv), so that (ii) against (iv) isolates what the issue adds downstream; (iii) seed plus path_issue, the best lexical ordering; (iv) seed plus rrf_pprpl_issue_path, the best fusion on the matched population and the one confirmed on held-out repositories; (v) seed plus rrf_hops_path, the best fusion by worst-case rank, carried as well because the two criteria select different fusions and we will not choose between them after the fact; (vi) seed plus rrf_pprpl_seedpath, the issue-free twin of (iv), so that the downstream

test inherits the control of Section 4.14; (vii) the repository truncated to the budget in a fixed canonical order, which is what a system with no retrieval step does; (viii) seed plus the key files themselves, cheapest first — the oracle ordering, an upper reference used to calibrate the budget, not a contender. Budgets of 16,000 and 64,000 tokens, chosen from Table 14 to sit either side of the point where most answers begin to fit.

Models. Two, of clearly different capability, so that the result is not a property of one model. A weak model that fails everywhere and a strong model that succeeds everywhere would both be uninformative; the pilot below exists to detect that before the full run.

Generation. A single API call per task and condition, with no tools and no retrieval available to the model. This is essential rather than incidental: a model that can search the repository defeats the budget, and the quantity under study is what a fixed context supports.

Outcome and analysis. A task is resolved if the generated patch applies and the specified tests pass. The comparison is paired across conditions on the same tasks, tested with McNemar’s test, Holm-corrected across the condition pairs. Because pull requests cluster within repositories, the primary analysis is a mixed-effects logistic regression with a random intercept per repository and condition as a fixed effect, with the McNemar result reported alongside; this follows the unit argument of Section 3.6 rather than the pull-request-level convention we abandoned there. We do not require the models to be reliable; we require the *difference between conditions* to be measurable, and the pairing makes the stochastic component common to all conditions.

Pilot and stopping rule. A 150-task pilot on the weaker model precedes the full run. If the oracle condition (viii) resolves under 10% or over 90% of pilot tasks, the budget is outside the informative range and is adjusted before the main run — and that adjustment is reported.

What would falsify the implication. If conditions (ii) to (vi) do not differ from (vii) at equal budget, then retrieval quality does not transfer to repair at these budgets, and the results of this paper should be read as being about retrieval alone. If (iv) does not beat (vi), the issue text does not earn its place downstream either, which would extend Section 4.14 from retrieval to repair. Both outcomes are publishable and we would report them.

7 THREATS TO VALIDITY

Construct. Neither evidence source is ground truth about what a developer needed to read; they are recoverable proxies with opposite biases, which is why we report both and never merge them. The co-edited key in particular conflates files that were *changed* with files that had to be *read*, and the symbol key is built by a resolver that follows import edges, so it is partly circular with the structural methods it favours; the human-annotated key of Section 4.11 confirms that its preference for structure does not carry over. That key has limits of its own: it covers 108 pull requests, most from SWE-bench Verified repositories that are heavily studied, and its annotators worked from the fix, so it records what the fix needed rather than what a developer consulted. Coverage per line assumes whole files are read, which overstates the cost for agents that read selectively; Section 4.8 shows which conclusions depend on that accounting. The execution-based source is absent (Section 3.3) and would be the natural third leg.

Internal. The graph builder resolves imports statically and misses dynamic imports and plugin registration. The symbol key is produced by a static analyser and inherits its resolution failures, and those failures depend on the environment the analyser runs in: rebuilding every key on a second machine left all 1,886 co-edited keys identical but changed the symbol key of 41 pull requests, usually by one file. Re-scoring with the rebuilt keys moves no AUC by more than 0.003,

swaps two adjacent orderings 0.0003 apart, and changes no conclusion; it does move some quoted figures in the third decimal, and the held-out counts of Section 4.9 to 194 and 191 of 200. The paper reports the keys as first built; the replication package records every key that differs and the re-scored comparison. Both biases favour structural methods. Issue-path leakage is measured and does substantially drive the co-edited result: it is reported in Section 4.13 as a finding rather than a reassurance, and the corresponding claim is stated conditionally throughout. Our leakage classifier is a regular expression over issue text and will mislabel some mentions in either direction; we report three nested strata, and three nested redaction arms, rather than one of each, so that the conclusion does not rest on a single cut. Leakage we have not thought to redact — a paraphrase of a function’s behaviour, a quoted error message defined in a key file — would sit in the residual that Section 4.14 does not attribute to leakage, so the leakage share is bounded below more firmly than above.

External. Python only, file granularity only, and pull requests drawn from benchmark suites that select for reproducible test-verified fixes — a population plausibly better documented than average, which may flatter the issue-text methods. The median repository here has 131 graph files and very large repositories are under-represented. The size split in Section 4.16 runs on the uncontrolled corpus, so it shows only that the pattern of Table 5 is stable across size quartiles rather than widening with size; it is not a size effect, but we cannot show the bias is harmless.

Conclusion. Pull-request-level aggregation, repository-level paired tests, Holm correction within comparison families and bootstrapping over repositories guard against the main inflation risks. Effect sizes accompany every test because several differences are significant and small. The fusion family, the worst-rank criterion and the directed walks were all added after the design freeze, so they are exploratory in origin; Section 4.9 shows the selection is stable across resamples of our repositories, and Section 4.10 that it holds, at the margin we allowed, on seven repositories outside the sample under predictions fixed in advance. Neither addresses the fact that the family was chosen after seeing results, and seven repositories are a small external set: one of the four predictions failed on it.

8 CONCLUSION

Given the file being changed, what else belongs in an agent’s context? Import-graph proximity and issue-text matching answer differently, and under two defensible definitions of the answer they appear to trade places. That reversal is mostly the comparison, not the signals: one half is the different pull requests the two keys cover, the other the issues that name their files.

What remains is two findings. Issue-aware retrieval is worth far more when the issue names a file the change touches, and between 0.061 and 0.078 of that 0.140 advantage is the naming itself; a benchmark average over issues that do and do not name their files describes neither, and should be reported per regime. And reciprocal-rank fusion of a length-aware walk with issue-to-path matching is the robust choice — never significantly worse than the best single signal on the pull requests carrying both keys, stable across repository splits, and holding on seven repositories outside the sample — except for single-file changes, where the walk alone is better. It costs a BM25 index over file paths and contents.

We recommend that context-retrieval work for code change report per evidence source, report and stratify by whether the query names the answer, compare any issue-aware method against the same method with a seed-derived query, match populations before comparing across keys, cluster its tests by repository, and use a repository prior rather than random order as its floor.

DATA AVAILABILITY

The replication package contains the import graphs at every base commit, the task definitions and evidence keys, the issue-token bags, the per-task orderings and coverage curves, the per-file token counts, and the analysis scripts that produce every number and figure in this paper; the external and ContextBench trees with their pre-registrations, which the version history shows were committed before the data they concern were scored; and the script that computes the dense embeddings, which are regenerated rather than shipped. A checker recomputes every number quoted in the text from the result files and fails if any sentence disagrees with them. It is available to reviewers at submission, and will be archived at a permanent identifier on acceptance.

USE OF AI TOOLS

Claude (Anthropic) was used to write the analysis and method code, to run the experiments and to draft this manuscript. All numeric results are produced by scripts in the replication package and were copied mechanically from their output files; the author verified the design, the claims and the text. Bibliographic entries were drafted with AI assistance and then checked against arXiv, the ACL Anthology and the publisher listings on 2026-09-23; the evaluation section of every system in Table 1 was read directly rather than summarised from an abstract. That check corrected eight entries drafted from memory, including a journal name, an author order, three titles changed between preprint versions, and one volume recorded as authored that is in fact edited. The corrections are listed in the replication package.

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A CORRECTION FAMILIES AND MIXED-MODEL ESTIMATES

Table 21. Every Holm family in the paper. A family is fixed by the design — one key, one unit, one population, one leakage stratum — rather than chosen per claim. Paired families hold all 325 tests among twenty-six orderings (the twenty contenders, two issue-free twins and four redacted variants); “keys” is how many such families exist in that row, one per key source (the union key is scored only on the full, unstratified corpus).

Tests	Population	Stratum	Keys	Tests per family
paired tests, pull request as unit	full	all	3	325
paired tests, pull request as unit	matched	all	3	325
paired tests, repository as unit	full	all	3	325
paired tests, repository as unit	full	explicit	2	325
paired tests, repository as unit	full	no explicit	2	325
paired tests, repository as unit	full	no full path	2	325
paired tests, repository as unit	full	no stem	2	325
paired tests, repository as unit	matched	all	3	325
paired tests, repository as unit	matched	explicit	2	325
paired tests, repository as unit	matched	no explicit	2	325
paired tests, repository as unit	matched	no full path	2	325
paired tests, repository as unit	matched	no stem	2	325
redaction, names arm	matched	by group	1	24
redaction, paths arm	matched	by group	1	24
redaction, symbols arm	matched	by group	1	24
mixed model, headline comparisons	both	several	1	41

Table 22. The comparisons claims rest on, re-estimated as a linear mixed model of the per-pull-request paired difference $a - b$ with a random intercept per repository. Positive means a is better; for a redacted twin, the difference is what the deleted names are worth. Holm-corrected as one family.

a	b	Key	Stratum	Pop.	PRs	Estimate [95% CI]	p_{Holm}
rrf_pprpl_issue_path	path_issue	co_edited	all	full	708	+0.053 [0.040,0.067]	$< 10^{-12}$
rrf_pprpl_issue_path	path_issue	co_edited	all	matched	702	+0.054 [0.040,0.067]	$< 10^{-12}$
rrf_pprpl_issue_path	path_issue	symbol	all	full	1156	+0.089 [0.075,0.104]	$< 10^{-30}$
rrf_pprpl_issue_path	path_issue	symbol	all	matched	702	+0.067 [0.054,0.079]	$< 10^{-23}$
ppr_out_pl	path_issue	co_edited	all	full	708	-0.105 [-0.125,-0.085]	$< 10^{-23}$
ppr_out_pl	path_issue	co_edited	all	matched	702	-0.106 [-0.126,-0.087]	$< 10^{-23}$
ppr_out_pl	path_issue	symbol	all	full	1156	+0.105 [0.082,0.127]	$< 10^{-17}$
ppr_out_pl	path_issue	symbol	all	matched	702	+0.012 [-0.012,0.037]	1
ppr_und_pl	path_issue	co_edited	all	full	708	-0.062 [-0.082,-0.042]	$< 10^{-7}$
ppr_und_pl	path_issue	co_edited	all	matched	702	-0.062 [-0.082,-0.043]	$< 10^{-7}$
ppr_und_pl	path_issue	symbol	all	full	1156	+0.103 [0.078,0.128]	$< 10^{-13}$
ppr_und_pl	path_issue	symbol	all	matched	702	+0.032 [0.007,0.056]	0.143
hops_lines	path_issue	co_edited	all	full	708	-0.044 [-0.064,-0.023]	$< 10^{-3}$
hops_lines	path_issue	co_edited	all	matched	702	-0.044 [-0.065,-0.024]	$< 10^{-3}$
hops_lines	path_issue	symbol	all	full	1156	+0.091 [0.067,0.115]	$< 10^{-11}$
hops_lines	path_issue	symbol	all	matched	702	+0.022 [-0.000,0.044]	0.610
rrf_pprpl_issue_path	ppr_out_pl	symbol	all	full	1156	-0.014 [-0.036,0.007]	1
rrf_pprpl_issue_path	ppr_out_pl	symbol	all	matched	702	+0.054 [0.032,0.076]	$< 10^{-4}$
rrf_pprpl_issue_path	ppr_out_pl	symbol	no explicit	full	871	-0.061 [-0.083,-0.040]	$< 10^{-6}$
rrf_pprpl_issue_path	ppr_out_pl	symbol	no explicit	matched	531	+0.000 [-0.024,0.024]	1
rrf_pprpl_issue_path	ppr_und_pl	symbol	all	full	1156	-0.013 [-0.036,0.009]	1
rrf_pprpl_issue_path	ppr_und_pl	symbol	all	matched	702	+0.035 [0.015,0.055]	0.008
rrf_pprpl_issue_path	ppr_und_pl	symbol	no explicit	full	871	-0.057 [-0.079,-0.035]	$< 10^{-5}$
rrf_pprpl_issue_path	ppr_und_pl	symbol	no explicit	matched	531	-0.016 [-0.037,0.005]	1
rrf_pprpl_issue_path	hops_lines	symbol	all	full	1156	-0.002 [-0.019,0.014]	1
rrf_pprpl_issue_path	hops_lines	symbol	all	matched	702	+0.045 [0.027,0.062]	$< 10^{-4}$
rrf_pprpl_issue_path	hops_lines	symbol	no explicit	full	871	-0.047 [-0.063,-0.032]	$< 10^{-7}$
rrf_pprpl_issue_path	hops_lines	symbol	no explicit	matched	531	-0.007 [-0.026,0.013]	1
hops_lines	path_issue	co_edited	no full path	matched	655	-0.016 [-0.037,0.004]	1
hops_lines	path_issue	co_edited	no explicit	matched	531	+0.034 [0.009,0.059]	0.103
hops_lines	path_issue	co_edited	no stem	matched	487	+0.050 [0.024,0.077]	0.004
ppr_out_pl	ppr_in_pl	symbol	all	full	1156	+0.215 [0.192,0.238]	$< 10^{-71}$
ppr_out_pl	ppr_in_pl	symbol	all	matched	702	+0.123 [0.104,0.143]	$< 10^{-33}$
rrf_pprpl_issue_path	rrf_pprpl_issue_path_rd	co_edited	explicit	matched	378	+0.063 [0.050,0.075]	$< 10^{-20}$
rrf_pprpl_issue_path	rrf_pprpl_issue_path_rd	co_edited	no explicit	matched	531	-0.002 [-0.005,0.001]	1
rrf_pprpl_issue_path	rrf_pprpl_issue_path_rd	symbol	explicit	matched	378	+0.049 [0.040,0.058]	$< 10^{-23}$
rrf_pprpl_issue_path	rrf_pprpl_issue_path_rd	symbol	no explicit	matched	531	-0.000 [-0.003,0.003]	1
rrf_pprpl_issue_path	rrf_pprpl_seedpath	co_edited	explicit	matched	378	+0.153 [0.130,0.176]	$< 10^{-36}$
rrf_pprpl_issue_path	rrf_pprpl_seedpath	co_edited	no explicit	matched	531	+0.020 [0.005,0.035]	0.133
rrf_pprpl_issue_path	rrf_pprpl_seedpath	symbol	explicit	matched	378	+0.116 [0.095,0.137]	$< 10^{-25}$
rrf_pprpl_issue_path	rrf_pprpl_seedpath	symbol	no explicit	matched	531	+0.012 [-0.003,0.027]	1